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A rapid transfer of virions coated with heparan sulfate from the ECM to cell surface CD151 defines a step in the human papillomavirus infection cascade

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eLife Assessment

This **useful** study presents interesting observations on the potential importance of extracellular transport of human papillomaviruses along actin protrusions by retrograde flow. The focus on the events of HPV infection between ECM binding and keratinocyte-specific receptor binding is unique and interesting. However, the evidence supporting the conclusions is **incomplete**, and additional experimental support is needed. Because conclusions drawn regarding HS interactions are largely based on experiments using a single HS mAb, the specificity of this mAb needs to be described in more detail, either based on the literature or further experimentation.

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Abstract

Human Papillomaviruses (HPVs) are the underlying cause of several types of cancer though they are mostly known for their association with cervical carcinoma. Before establishing an infection, the virions must reach their target cells through a break in the epithelial barrier. After binding to heparan sulfate (HS) of the extracellular matrix (ECM), they translocate to the cell surface and co-internalize with the entry factor CD151.

For studying these early events of the infection cascade, we block the translocation from ECM-attachment sites to the cell body, release the block, and monitor the association of virions with CD151 or HS. We observe quick virion translocation from the ECM to the cell body within 15 min. During this process, virions associate with the tetraspanin CD151 present at the cell border or at filopodia. Translocating virions are decorated with HS, which they lose in the next few hours, presumably prior to endocytosis.

Our observations reveal a rapid step in the HPV infection cascade: the transfer of HS-coated virions from the ECM to CD151. This step is too fast to account for the asynchronous uptake of HPVs which is likely driven by glycan-processing and HS uncoating contributing to virus structural activation in preparation for endocytosis.

Introduction

Already in the 80s, Harald zur Hausen proposed a role of human papillomaviruses in cancer (zur Hausen, 2009). Since then, five more classes of oncogenic viruses have been identified (Galati et al., 2024a). To date, it is assumed that more than 10% of the worldwide human cancer burden is associated with infectious agents (Galati et al., 2024a), from which about a half is caused by *Papillomaviridae* (Martel et al., 2017) For this reason, the understanding of viral entry strategies has implications going beyond the classical treatment of acute viral infections.

Human papillomaviruses are small, non-enveloped viruses with a diameter of 55 nm. The icosahedral capsid is mainly composed of pentameric L1 capsomers. Together with fewer L2 capsid proteins, capsomers surround a histone core bearing a circular double-stranded DNA (Baker et al., 1991; Ozbun and Campos, 2021). From the more than 200 phylogenetically classified HPV genotypes, the most oncogenic ones are HPV16 and HPV18 (Galati et al., 2024b), which are responsible for about 70-80% of the cervical cancer cases (Christiansen et al., 2015). In addition, they cause other severe cancers such as anogenital, head and neck tumors (Doorbar et al., 2012).

Papillomavirus infection requires a break in the epithelial barrier, through which virions reach mitotically active basal cells of the epithelia (Ozbun and Campos, 2021). Here, virions bind to the linear polysaccharide heparan sulfate (HS) that is an important constituent of the extracellular matrix (ECM) and the plasma membrane. In the latter case, it is attached to membrane proteins, so called heparan sulfate proteoglycans (HSPGs). Positively charged and polar residues of the L1 capsid protein interact with negative charges of HS, resulting in a strong bond (Dasgupta et al., 2011; Giroglou et al., 2001; Joyce et al., 1999; Knappe et al., 2007; Surviladze et al., 2015). While in cell culture virions bind to HS of the cell surface and the ECM, it has been suggested that *in vivo* they bind predominantly to HS of the extracellular basement membrane (Day and Schelhaas, 2014; Kines et al., 2009; Schiller et al., 2010).

Regardless of where the virions bind to HS, they must detach from the linear polysaccharide before they bind to cell surface receptors and are internalized (Ozbun and Campos, 2021). However, the strong electrostatic bonds do not allow for HS-virion dissociation. Two release mechanisms are discussed, that mutually are not exclusive. In one model, binding to HS structurally results in capsid enlargement and softening (Feng et al., 2024), finally resulting in the exposure and cleavage of capsid proteins (Becker et al., 2018), which is required for further steps in infection. In an alternative model, the direct HS-virion bond remains. However, cleavage of HS/HSPGs by proteinases and heparanases produces fragmented HS that, albeit still bound to the virion surface, no longer immobilizes the virion (Surviladze et al., 2012; Surviladze et al., 2015).

Once the virions are released, they could reach the cell surface simply by passive diffusion. In a co-culture trans-well assay, PsVs from donor cells infect spatially separated receiver cells (Surviladze et al., 2012), demonstrating that free diffusion of virions is sufficient for infection. However, an active transport mechanism has been reported as well. PsVs migrate along actin-rich protrusions from the ECM towards the cell body (Schelhaas et al., 2008; Smith et al., 2008).

While it is generally agreed on that HS is a primary virion attachment site, the molecular identity of the receptor complex on the cell surface is unknown. This complex is most likely a multimeric complex rather than a single molecular component. Possible candidates are proteins crucial for cell entry, as the tetraspanin CD151 (Mikuličić et al., 2019; Scheffer et al., 2013; Spoden et al., 2008), integrin-α6 (Itgα6) (Evander et al., 1997; Yoon et al., 2001), growth factor receptors (Mikuličić et al., 2019; Surviladze et al., 2012), and the annexin A2 heterotetramer (Dziduszko and Ozbun, 2013; Woodham et al., 2012). From these molecules, the tetraspanin CD151 could play a coordinating role, as tetraspanins have been suggested to organize viral entry platforms in several types of viral infections, including infections with coronavirus, cytomegalovirus, hepatitis C virus, human immunodeficiency virus, human papilloma virus, and influenza virus (Bruening et al., 2018; Earnest et al., 2015; Florin and Lang, 2018; Hantak et al., 2019; Hochdorfer et al., 2016; Scheffer et al., 2013; Zona et al., 2013). Tetraspanin entry platforms could form in a slow and stochastic manner, providing an explanation for the asynchronous virion uptake with half-times of above 10 h (Becker et al., 2018). However, it is unclear whether virions associate with CD151 already during virion transfer from the ECM to the cell surface, or perhaps much later, in preparation of the endocytic uptake.

In this study, we explore the actin-dependent transport of HPV-PsV infection by employing the cell permeable mycotoxin and actin polymerization inhibitor cytochalasin D (CytD). We find that CytD causes the trapping of HPV16 pseudovirions (PsVs) in the ECM, where they remain accumulated adjacent to the cell periphery. Upon CytD removal, HS-decorated PsVs move from the ECM to cell surface CD151. The association of PsVs with CD151 persists within the next few hours, whereas the HS coat is stripped off. These findings distinguish a step in the infection cascade with CD151 playing an early role directly after the release of virions from the ECM.

Results

Cytochalasin D arrests the transfer of PsVs from the extracellular matrix to the cell surface

The molecular surface of PsVs is immunologically indistinguishable from HPV virus particles (Doorbar et al., 2012), which makes them a widely used tool for studying host cell entry. In this study, we employ HPV16-pseudovirions with an encapsidated luciferase reporter-plasmid under the control of the HPV16-promotor. Instead of viral-DNA, a luciferase-encoding plasmid enters the nucleus, enabling the analysis of the infection rate via the luciferase activity. Additionally, the plasmids are composed of nucleotides to which fluorophores can be coupled by click-chemistry. This allows for their detection by fluorescence microscopy throughout the entire infection pathway.

Many PsVs in the ECM are too far away from the cell surface to allow for direct physical contact between PsVs and entry receptors. In order to reach the cell surface, virions migrate by an actin-dependent transport along cell protrusions towards the cell body (Schelhaas et al., 2008; Smith et al., 2008). We aimed for blocking this transport in HaCaT cells, a human keratinocyte cell line that is widely used as a cell culture model for HPV infection. Cells are incubated for 5 h with PsVs in the absence or presence of 10 µg/ml CytD. Previously, it has been shown that this CytD concentration stops cell migration only after a few minutes (Peng et al., 2011). Therefore, we assume that CytD does not immediately stop PsV translocation. **Figure 1A** shows cells that, after the 5 h incubation, are fixed and stained by a membrane marker (the dye TMA-DPH) to mark the main cell body (gray), for PsVs (by click-chemistry; magenta), HS (antibody staining; cyan), and the membrane protein Itgα6 (antibody staining; green). In the control, PsVs frequently locate within the area of the cell body (**Figure 1A**, upper row; see outline based on TMA-DPH staining). In contrast, when CytD was present, the cell body is largely devoid of PsVs (**Figure 1A**, lower row; please note that the brightness in the center is mainly caused by autofluorescence). Instead of being transported to the cell body as in the control, PsVs accumulate adjacent to the cell body. This region is likely to be the ECM as it is up to several µm wide and rich in HS. We conclude that, upon inhibition of actin-dynamics by CytD, PsVs remain accumulated in the ECM, that in this assay is clearly distinguishable from the cell body.

Comparing the total PsV intensities in the upper (control) and lower (CytD) magenta panels in **Figure 1A** suggests that the amount of accumulated PsVs at the periphery does not differ greatly from the amount of PsVs in the control. This is surprising, as CytD should not inhibit enzymes involved in capsid- or HS/HSPGs-processing, which is supposedly sufficient for the release of PsVs. Therefore, we conclude that potentially releasable PsVs remain in association with the ECM or the cell body. However, a quantitative comparison is not possible in this experiment because of the central autofluorescence (**Figure 1A**, large magenta area in the center) overlapping with some of the PsVs. For that reason, we employed unroofed cells (Heuser, 2000), also called membrane sheets. After treatment of cells with PsVs and CytD as above, membrane sheets are generated by brief ultrasound pulses that remove the upper cellular parts (and thus any intracellular autofluorescence), leaving behind the extracellular matrix and the basal membrane along with the bound PsVs. During imaging, membrane sheets are identified via staining of F-actin with fluorescent labelled phalloidin (**Figure 1B**, green). As in the experiment above (**Figure 1A**), PsVs remain accumulated after CytD treatment, adjacent to the now isolated basal membrane (**Figure 1B**, magenta).

The PsV maxima density (**Figure 1B**, magenta, detected as local maxima of antibody stained L1) and the PsV intensity are quantified in an area covering the cell body and the periphery. Compared to the control, CytD reduces the PsV maxima density by~30% (the PsV maxima density in the control and CytD treated cells is 0.7 and 0.5 PsVs/µm^2, respectively), whereas the maxima intensity increases by~40% (**Figure 1C**). A histogram of the PsV maxima intensities illustrates that CytD broadens the intensity distribution towards many PsVs that are several-fold brighter than the most frequent maxima intensity of the control (**Figure 1D**). Brighter PsVs resulting from CytD treatment can also be seen in the L1 images of **Figure 1B** (shown at the same scaling). We assume that this observation is due to the limited microscopic resolution, as CytD is unlikely to act directly on PsVs (e.g. by fusing them to larger and brighter particles). The accumulation of PsVs results in overlapping and thus poorly resolved maxima which are~40% brighter but of which there are~30% fewer. The decrease in maxima density is in the range of the increase in intensity, which yields a similar total signal in both conditions.

We addressed whether PsVs would still be accumulated when adding CytD 1 h after adding the PsVs rather than simultaneously (**Figure 1B**, lower panel, CytD after 1 h). As in the CytD condition, we see accumulated PsVs adjacent to the isolated basal membrane (**Figure 1B**, arrows point towards PsV accumulations). Furthermore, we find a broader PsV maxima intensity distribution compared to the control but a narrower intensity distribution compared to the sample where CytD is present during the entire 5 h incubation (**Figure 1D**). We conclude that without CytD it takes longer than 1 h for the PsVs to be translocated to the cell body. However, the preparation of PsVs for the translocation step likely is completed after 5 h.

Hence, it appears that CytD only arrests PsVs on their journey towards the cell surface. However, our assay is not very precise, and we do not know how many PsVs in the control were already endocytosed, which would reduce the reference value. Therefore, we can only say that within the 5 h incubation period we do not lose a major part of accumulated PsVs after processing in the ECM.

The blocking of PsV endocytosis by cytochalasin D is reversible

CytD not only arrests the transport of PsVs to the cell but is known to block other actin-dependent processes, which strongly affects the physiology of the cell. Therefore, we investigated whether PsVs would proceed normally on their infection pathway, after removal of CytD. In order to allow

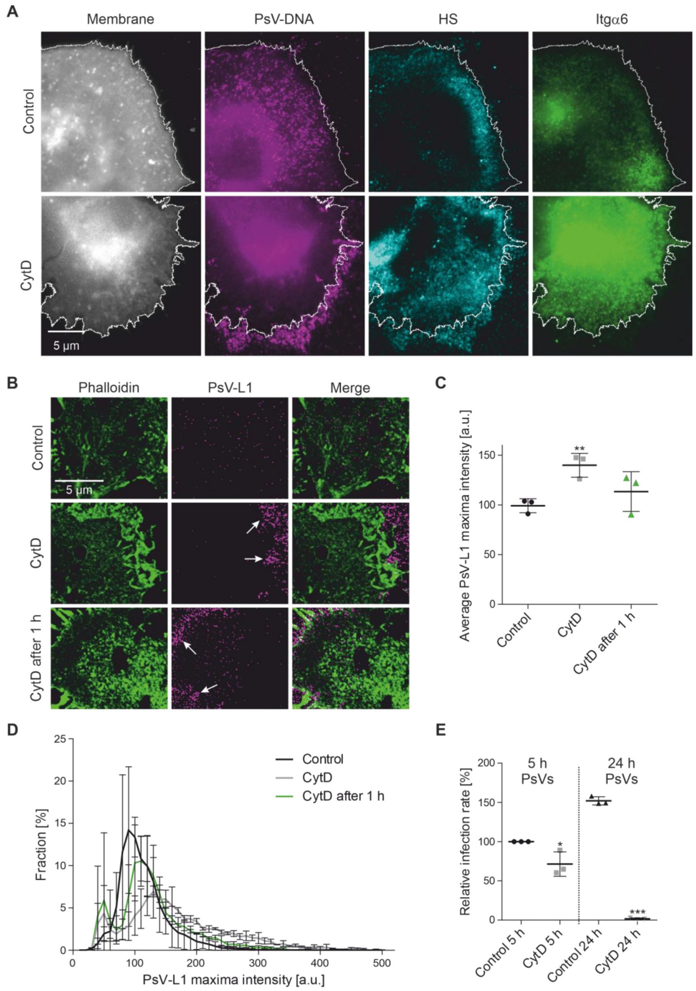


Figure 1.

CytD arrests PsV translocation from the ECM to the cell body.

(A) In the absence (control) or presence of 10 µg/ml CytD, HaCaT cells are incubated with PsVs at 37 °C for 5 h. Then, cells are fixed and stained with the cell membrane dye TMA-DPH (grey). PsVs (magenta) are visualized through coupling a dye (6-FAM Azide) to the encapsidated plasmid by click-chemistry, and indirect immunolabeling is used for staining of HS (AlexaFluor™ 594; cyan) and Itgα6 (STARRED; green). Imaging was realized with epi-fluorescence microscopy. White lines mark the main cell body; they are generated with reference to the membrane staining. (B) Same pre-treatment of cells as in (A) with an additional condition where CytD is added 1 h after the PsVs (CytD after 1 h). Prior to fixation, membrane sheets are generated and F-actin is stained with phalloidin iFluor488 (green) and PsVs are stained by immunofluorescence using an antibody against L1 in combination with an AlexaFluor™ 594-labelled secondary antibody (magenta; not shown in this figure for clarity reasons is an additional staining of CD151 with STAR RED). Images of phalloidin and L1 are acquired in the confocal and STED mode of a STED microscope, respectively. Arrows in the magenta panels point towards accumulated PsVs. (C) Images as shown in (B) are analyzed. PsV maxima are detected and their intensities are quantified in a circular 125 nm diameter region of interest (ROI), followed by background correction. Values are given as means ± SD (n = 3 biological replicates; one biological replicate includes per condition 14 - 15 analyzed membranes with altogether at least 1000 maxima). (D) PsV maxima intensity distribution of the data in (C). The fraction of PsVs, expressed in percent, is plotted against the maxima intensity. Values are given as means ± SD (n = 3). (E) HaCaT cells are treated either for 5 h or 24 h with PsVs, with or without 10 µg/ml CytD. In case of the 5 h incubation, cells are washed and incubated for another 19 h in medium (in total 24 h). After a total of 24 h incubation, the luciferase activity of lysed cells is measured, yielding the infection rate that is normalized to LDH, resulting in the normalized infection rate. The normalized infection rate is further related to the mean normalized infection rate of the 5 h control, set to 100 %, yielding the relative infection rate. Values are given as means ± SD (n = 3). (D) and (E). Statistical differences between control and CytD is analyzed by using the two-tailed, unpaired student's *t* test (n = 3). a.u., arbitrary units.

for recovery from CytD, cells were washed after the 5 h PsV/CytD treatment, then incubated for another 19 h and the infection rate was subsequently determined by measuring the luciferase activity in the cell lysate. Compared to the control, CytD reduced luciferase activity by 30% (**Figure 1E**; left). A certain degree of reduced infectivity is to be expected, as PsVs have ~20% less time to complete infection as compared to PsVs infecting cells that were not treated with CytD. When PsVs were not washed off and incubated for 24 h (5 h + 19 h), the infection rate increased by 50 %, whereas continuous treatment with CytD results in no infection (**Figure 1E**; right). The latter is in line with previous studies (Schelhaas et al., 2008; Selinka et al., 2002; Selinka et al., 2007; Spoden et al., 2013) showing that CytD is a strong inhibitor of HPV infection. In any case, PsVs proceed on the infection pathway when CytD is removed. The reduced infection rate of 30% can be explained, at least in part, by the 5 h delay by which PsVs reach the cell surface. Therefore, we propose that CytD is suitable for transiently arresting PsVs in a state between primary attachment to HS and cell surface binding.

The transfer of PsVs from the ECM to the cell body is fast and efficient

Next, we studied the time course of PsV translocation. Cells were treated as in **Figure 1A** (in **Figure 2** the 0 min time point is identical), followed by washing off of PsVs/CytD and fixation after 0, 15, 30 and 60 min. The integrated density of PsVs locating at the cell periphery (**Figure 2A**, the cell periphery is defined by the white lines) is quantified and plotted against time. Compared to the control, CytD treatment results in a 6-fold increase of PsVs in the periphery when cells are immediately fixed after washing off of CytD (0 min time point). This increase is more than halved when cells were incubated for another 15 min after washing off of CytD and after 30 min, the level of the control is reached (**Figure 2B**). Hence, the half-time of PsV translocation from the periphery to the cell body is about 15 min, demonstrating that translocation is fast and efficient. Considering the fact that prior to endocytosis virions remain several hours on the cell surface, the duration of translocation is short, making it unlikely that it contributes to the asynchronous virion uptake that is observed for HPV-PsVs.

Translocation of PsVs to CD151

As shown above, after CytD removal, the accumulated PsVs approach the cell surface. During this process, they may be coated with ECM cleavage products, and at some point, bind to cell surface HPV entry factors, like the tetraspanin CD151. For studying these events in detail, we employ superresolution STED microscopy, analyzing the association of PsVs with HS and CD151 over time. Cells are treated and monitored as in **Figure 2**, but with an extended time window of up to 180 min, as cell surface processes are expected to take longer than an hour.

After fixation, PsVs and CD151 were double-stained with antibodies against L1 and CD151, without prior cell permeabilization (cell surface staining, albeit fixation perforates the cell membrane to some extent). In addition, we stained F-actin with fluorescent labelled phalloidin. We simultaneously image L1 and F-actin in the confocal and CD151 in the STED channel.

As shown in **Figure 3A**, CD151 concentrates in spots scattered across the cell surface. It is also present at cell protrusions that are rich in F-actin and vary strongly in number and shape (**Figure 4A** shows anecdotal images of these filopodia). The antibody staining against PsVs, like the PsV click-chemistry staining in **Figure 1A** and **2A**, is highly variable. CytD conserves the accumulated PsVs at the cell periphery and at early time points PsVs appear brighter (**Figure 3A**). Similar to **Figure 2**, we analyzed the time course of the diminishment of accumulated PsVs but focused on PsVs that are very close to or have already approached the cell surface (**Figure 4B** and **C**, for details see Figure legend of 4B). Moreover, we counted the number of PsVs instead of measuring the integrated density. As shown in **Figure 4C**, the PsVs accumulated in the cell border region diminish with about the same rate as PsVs in the periphery (**Figure 2B**).

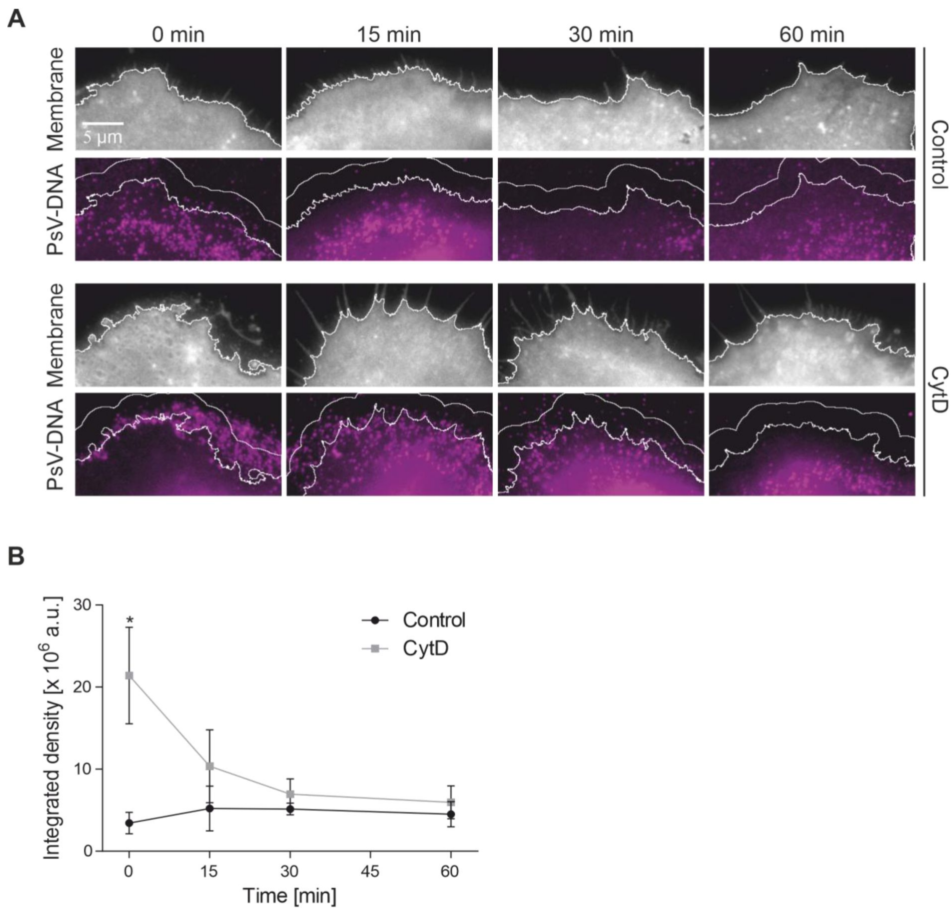


Figure 2.

Fast diminishment of the accumulated PsV signal from the cell periphery after washing off of CytD.

(A) HaCaT cells are incubated with PsVs at 37 °C for 5 h, in the absence (control) or presence of CytD (10 µg/ml). Then, cells are washed and incubated for the indicated time periods without PsVs/CytD, before they are fixed and stained as in **Figure 1** (t = 0 min is identical to **Figure 1A**; for clarity we show only the membrane (grey) and the PsV staining (magenta)). White lines in the membrane images are generated with reference to the membrane staining. One line marks the cell body. The cell body line was broadened by 30 pixels (see additional smoother white lines in the PsV channel, magenta). (B) Magenta images, the area enclosed by the two white lines mark the cell periphery. The PsV signal of the periphery is quantified as integrated density, background corrected, and plotted over time. Values are given as means ± SD. The statistical difference between the same time points of control and CytD is analyzed by using the two-tailed, unpaired student's *t* test (n = 3 biological replicates).

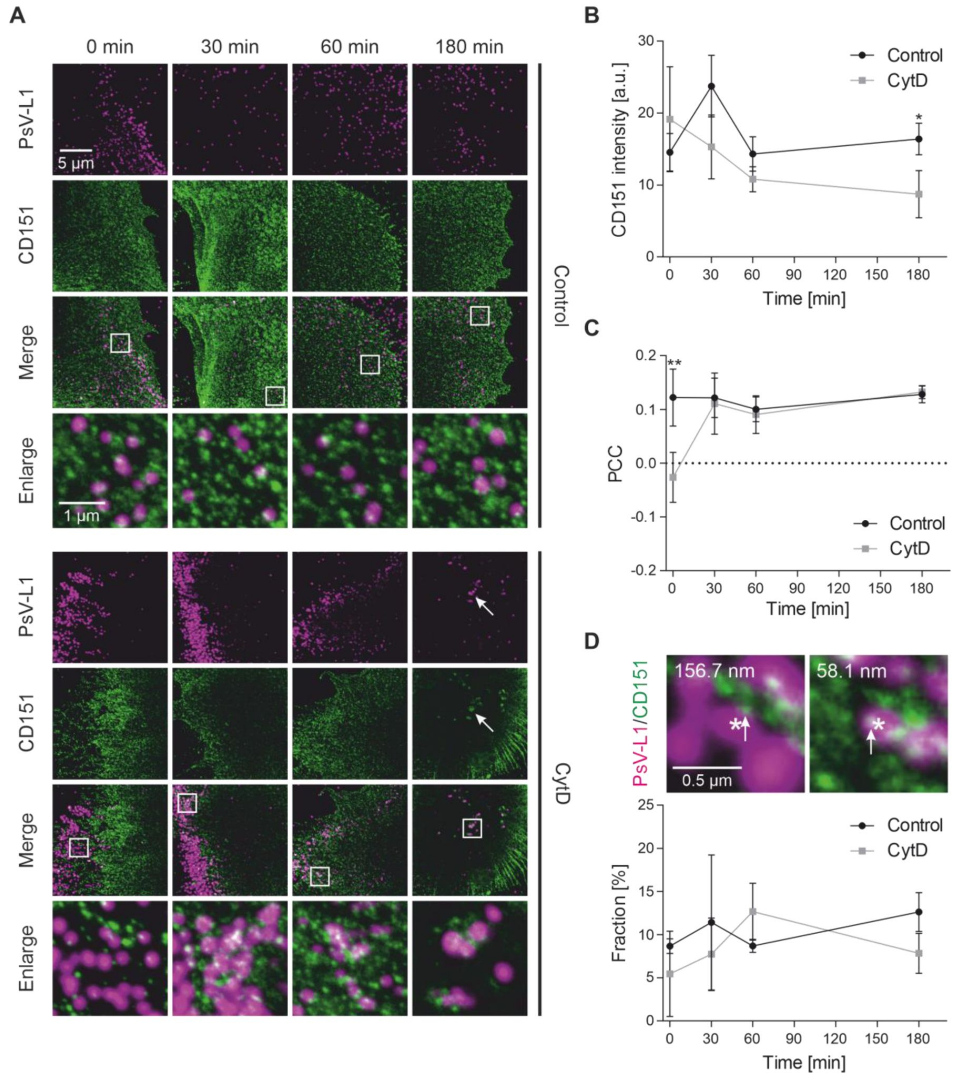


Figure 3.

Association between PsVs and CD151.

(A) HaCaT cells are incubated with PsVs at 37 °C for 5 h, in the absence (control, upper panels) or presence of 10 µg/ml CytD (CytD, lower panels). Afterwards, cells are washed and incubated without PsVs/CytD further for 0 min, 30 min, 60 min or 180 min, before they are fixed and stained by indirect immunofluorescence for L1 (magenta; STAR GREEN) and for CD151 (green; AlexaFluor™ 594), and for F-actin by fluorescent labelled phalloidin (iFluor647; here not shown for clarity, please see **Figure 4** for the F-actin staining). The bottom rows show enlarged views of the merged images, from the regions marked by the white boxes. PsVs (L1 staining) and F-actin are imaged in the confocal and CD151 in the STED mode of a STED microscope, respectively (see enlarged views). 180 min/CytD, arrows mark presumably endocytic structures in the central cell body region (for more examples see also **Supplementary Figure 1**). For analysis, we place rectangular ROIs onto the images that cover mainly the cell body but include parts of the cell periphery as well (see example in **Supplementary Figure 2A**). (B) Within these ROIs, the average CD151 intensity is measured and plotted over time. In the same ROIs, (C) the Pearson correlation coefficient (PCC) between PsV-L1 (magenta) and CD151 (green) is calculated and plotted over time. (D) the fraction of PsVs (in percent) that have a distance to the next neighbored CD151 maximum ≤ 80 nm, which we define as tightly associated, is plotted over time. The fraction of PsVs tightly associating with CD151 is corrected for random background association (for details see **Supplementary Figure 3**). (D) Two examples of PsVs (each marked by an asterisk) taken from the 30 min/CytD (left) and 60 min/CytD (right) conditions. Number in the upper left, the shortest distance between the PsV-L1 maximum and the next nearest CD151 maximum (marked by an arrow) is given in nm. Values are given as means ± SD (n = 3 biological replicates). Statistical difference between the same time points of control and CytD is analyzed by using the two-tailed, unpaired student's *t* test (n = 3).

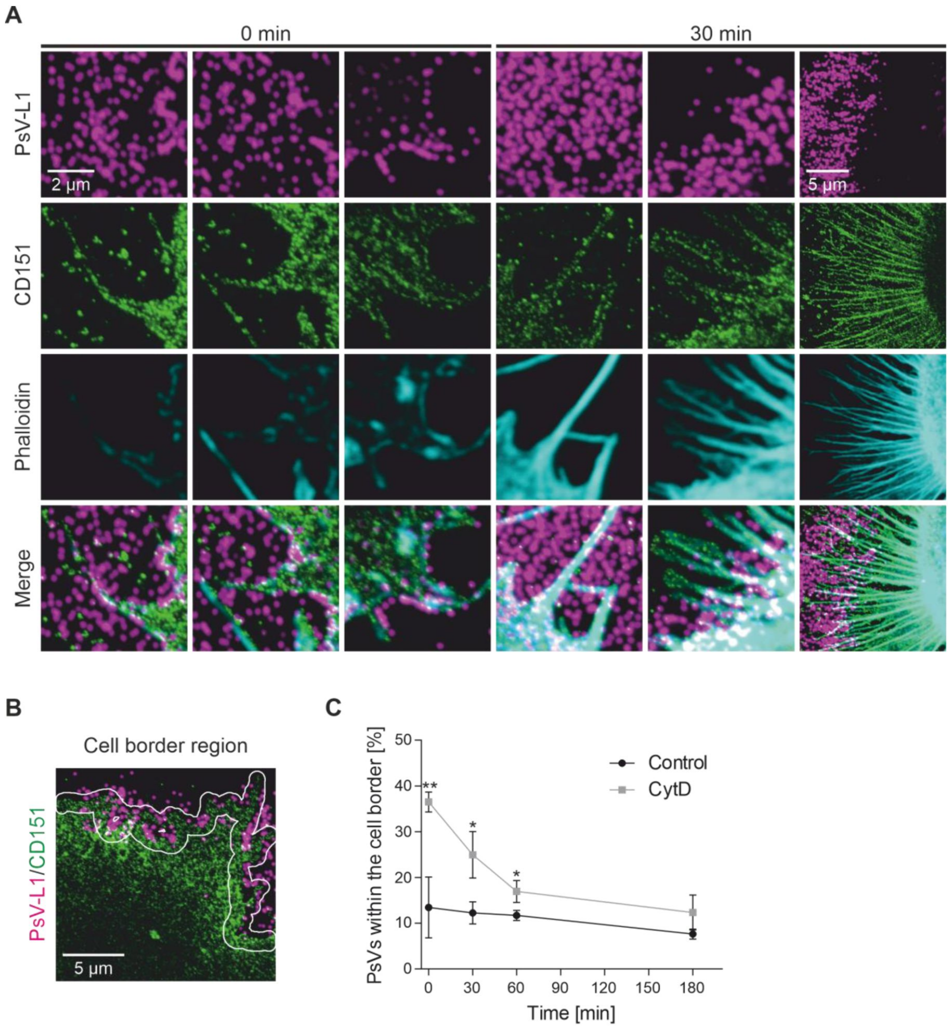


Figure 4.

Gallery of images illustrating the variability of filopodia/the diminishment of the number of PsVs at the cell border region after CytD removal.

(A) Images are taken from the same experiment described in **Figure 3**. Many cells have filopodia. However, due to the large variability in number and shape it is impossible to show representative images. Therefore, the shown examples taken from the CytD condition are anecdotal images of cells with CD151 positive filopodia. (B) Based on the CD151 image, a cell border region is broadened to 40 pixels. Please note that this cell border region is different from the cell periphery described in **Figure 2**. In this analysis, the analyzed region covers both sides of the cell border (approximately 75% inside and 25% outside of the cell; for details see methods), and therefore is referred to as the cell border region. (C) Diminishment of PsV maxima from the cell border region over time, expressed as percentage of all PsVs present in the image. Values are given as means ± SD (n = 3). Statistical difference between the same time points of control and CytD is analyzed by using the two-tailed, unpaired student's *t* test (n = 3).

Just like the L1 staining of PsVs, the intensity of the CD151 staining between cells is highly variable (**Figure 3A** and **B**). In the control, the mean intensity does not change over the 180 min time course, whereas a trend towards a lower mean intensity is observed in the cells treated with CytD, with the last time point being significantly lower than the control (**Figure 3B**). The decrease of the CD151 staining intensity points towards the possibility that after CytD wash off some CD151 is internalized, presumably due to co-internalization with endocytosed PsVs. This idea is supported by the observation that in particular at 180 min/CytD we observe occasionally CD151/PsV agglomerations (**Figure 3A**, see structures marked by arrows at 180 min/CytD, for more examples see **Supplementary Figure 1A**). We did not study this issue systematically, but some of these structures have clear three-dimensional extension (see **Supplementary Figure 1B** for axial scans) and therefore likely are tubular structures filled with several PsVs, as previously described by electron microscopy (Schelhaas et al., 2012). We found fewer of such structures at 180 min/control, as cells might have been actively interacting with PsVs for altogether 8 h, opposed to 3 h in the CytD treated cells. Hence, in the control, PsV endocytosis is less synchronized compared to CytD and therefore we do not find a notable CD151 diminishment.

For studying the association between PsVs (L1) and CD151, the Pearson correlation coefficient (PCC) between the channels is calculated. In theory, the PCC would equal 1 in the case of a perfect overlap and −1 for an image and its negative. We should not obtain large PCC values, as in most images PsVs are less abundant than CD151. As expected, the PCCs are around 0.1, with the exception of the 0 min/CytD value that is significantly lower and even slightly negative (**Figure 3C**, for control PCC values of flipped images see **Supplementary Figure 2**). This reflects the partially mutual exclusion of the two stainings; PsV accumulations are at the cell periphery and the CD151 staining is mainly at the cell body. At 30 min/CytD, the PCC has reached the level of 0.1 as in the other conditions, in line with **Figure 2B** showing that the accumulated PsVs diminish within 15 min.

PsV translocation to the cell surface should increase the number of PsVs that are tightly associated with CD151. As criteria for tight association, we define a distance of 80 nm between PsV and CD151 maxima, a value which is close to the resolution limit of the used microscope (Finke et al., 2020). In the control, the fraction of PsVs tightly associated with CD151, after correction for random background association (see **Supplementary Figure 3**) is about 10% (**Figure 3D**, control). At 0 min/CytD, we start with a fraction of 5.5%, a value that more than doubles in the next 60 min, although these changes are not significant (**Figure 3D**, CytD).

In summary, we conclude that within 180 min after a 5 h pre-incubation with PsVs (**Figure 3D**, control) around 10% of the PsVs associate tightly with CD151, with no trend towards more/less association over time. With the exception of the lower 0 min/CytD value of 5.5%, CytD has no effect on the association (**Figure 3D**, CytD). The 0 min/CytD value could be an underestimation, because (i) PsVs are optically less well-resolved (which increases the distance of a PsV to CD151) and (ii) this time point in particular is likely to be overcorrected for background association (see legend of **Supplementary Figure 3**). Despite of these uncertainties, the data shows that PsVs establish contact to CD151 assemblies either while still in the ECM, or in the moment of leaving the ECM.

PsV association with HS

Next, we studied the association between PsVs and HS, including a reference staining for the cell body as in **Figure 1**. However, we cannot image TMA-DPH with our STED microscope. Instead of TMA-DPH, we used Itgα6 as a reference marker for the cell body as it is not detected at cell protrusions (compare TMA-DPH and Itgα6 in **Figure 1A**). PsVs are visualized by click-chemistry and imaged in the confocal channel. HS and Itgα6 were stained by antibodies without prior permeabilization, and imaged at STED microscopic resolution. The three stainings were simultaneously recorded.

As shown in **Figure 5A** (green), the Itgα6 staining results in a densely spotted pattern. The Itgα6 intensity does not change over time (**Supplementary Figure 4E**). The pattern of the HS staining is variable regarding the overlap of HS with both, PsVs and Itgα6 (**Figure 5A**). Moreover, CytD treatment increases the HS intensity (**Figure 5B**, also visual when comparing the brightness of HS in **Figure 5A** upper and lower images). This increase of intensity is particularly notable at the 0 min time point, where the samples treated with CytD have a more than two-fold higher intensity and differ significantly from the control. This suggests that CytD prevents loss of HS occurring during or after PsV translocation.

In the control, the PCC between PsVs and HS is close to zero (**Figure 5C**, for control PCC values of flipped images see **Supplementary Figure 4**). The largest PCC is found at 0 min/CytD, which reflects the finding that at this time point both PsVs and HS preferentially locate at the cell periphery. However, over time the accumulated PsVs diminish by translocating to the cell surface, which is accompanied by the PCC decreasing to zero, the same value as in the control (**Figure 5C**). Additionally, we analyze the PCC between PsVs and HS specifically in the cell body region, excluding the cell periphery (**Figure 5D**). The PCC in the cell body region is close to zero for the control. After CytD, we observe a trend towards a higher PCC between 0 min and 30 min. The fraction of PsVs tightly associating with HS at 0 min is increased by more than 3-fold by the CytD treatment. Over the next 180 min, the fraction decreases until it reaches the control value (**Figure 5E**).

Altogether, the analysis of the PCC between HS and PsVs (**Figure 5C**) and the fraction of tightly HS associated PsVs (**Figure 5E**) shows that CytD increases at 0 min the HS/PsV association, because PsVs remain accumulated in the ECM. After CytD removal, this association is lost, only in the region of the cell body is a trend towards a higher PCC at 30 min (**Figure 5D**) that could reflect HS-coated virions translocating to the cell surface.

Next, we plot for each PsV their distance to the next nearest Itgα6 maximum against the distance to their next nearest HS maximum. In the control, the distances remain unchanged over the entire observation time; 59% - 65% of the PsVs have a short distance (< 250 nm) to both Itgα6 and HS (**Figure 6B**, black; see also **Supplementary Table 1**), and another 12 - 15% a short distance (< 250 nm) to HS but not to Itgα6 (> 250 nm) (**Figure 6B**, green). The distance plots shown in **Figure 6C** are in line with the observation that the distance patterns of PsVs in untreated cells do not change over the 180 min observation time (**Figure 6C**, please compare the upper distance plots from left to right). In cells treated with CytD, a larger fraction of PsVs is accumulated at the periphery at the 0 min time point, as marked by a larger fraction (46% as opposed to 12% in the control; **Figure 6B**, green) of PsVs with a short distance (< 250 nm) to HS and a large distance (> 250 nm) to Itgα6. Over time, the PsVs in the CytD treated cells (**Figure 6C**, lower row) acquire a shorter distance to Itgα6 and a larger distance to HS. After 180 min, the distances are similar to the untreated control.

If HS would bind irreversibly to PsVs, the population with short HS-PsV distances is expected to be constant. However, we observe that this population diminishes. This can only be explained by HS uncoating. In fact, between 60 min/CytD and 180 min/CytD, the fraction of PsVs with a large distance (> 250 nm) to HS and a short distance (< 250 nm) to Itgα6, representing PsVs at the cell body without HS, increases from 9.6% to 25.2% (**Figure 6B**, pink). The observation that PsVs lose their HS coat after translocating to the cell surface is also supported by the lower HS intensity in the control (**Figure 5B**) and the transient increase in the PCC between HS and PsVs in the cell body at 30 min (**Figure 5D**).

The movement of HS towards the cell body is demonstrated by a shortening of the HS-Itgα6 distance over time (**Supplementary Figure 4D**). Moreover, the PCC between PsVs and HS in the cell body area increases (**Figure 5D**). Altogether, the data suggest that HS-coated PsVs migrate towards the cell body. A majority of translocating PsVs shed HS within one hour after washing off

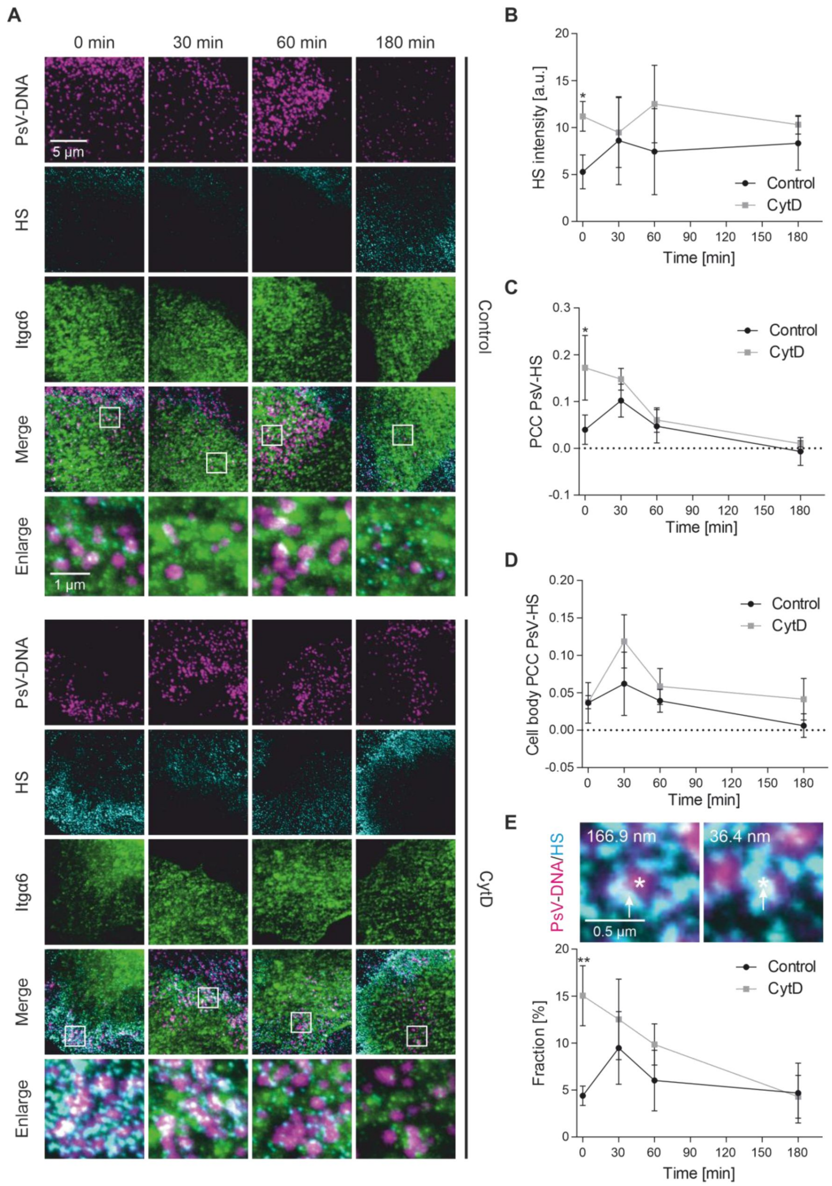


Figure 5.

Association between PsVs and HS.

(A) HaCaT cells are incubated with PsVs at 37 °C for 5 h, in the absence (control, upper panels) or presence of 10 µg/ml CytD (CytD, lower panels). Afterwards, cells are washed and incubated without PsVs/CytD further for up to 180 min, before they are fixed and stained. PsVs (magenta) are visualized by click-chemistry (6-FAM Azide, see also above) and indirect immunolabeling is used for HS (cyan; AlexaFluor™ 594) and for Itgα6 (green; STAR RED). Shown in the bottom rows are enlarged views of the white boxes in the merged images. PsVs (DNA staining) are imaged in the confocal and HS and Itgα6 in the STED mode of a STED microscope, respectively (see enlarged views). For analysis, we place rectangular ROIs onto the images that cover mainly the cell body but include parts of the cell periphery as well (see example in **Supplementary Figure 4A**) (B) Average HS intensity over time. (C) PCC between PsV-DNA (magenta) and HS (cyan) over time. (D) PCC between PsV-DNA (magenta) and HS (cyan) in the region of the cell body over time. (E) The fraction of PsVs (in percent) tightly associating with HS (distance ≤ 80 nm) plotted over time (for background correction see **Supplementary Figure 5**). Two examples of PsVs (each marked by an asterisk) taken from the 0 min/CytD condition. Number in the upper left is the shortest distance in nm between the PsV-DNA maximum and the next nearest HS maximum (marked by an arrow). Values are given as means ± SD (n = 3 biological replicates). Statistical difference between the same time points of control and CytD is analyzed by using the two-tailed, unpaired student's *t* test (n = 3).

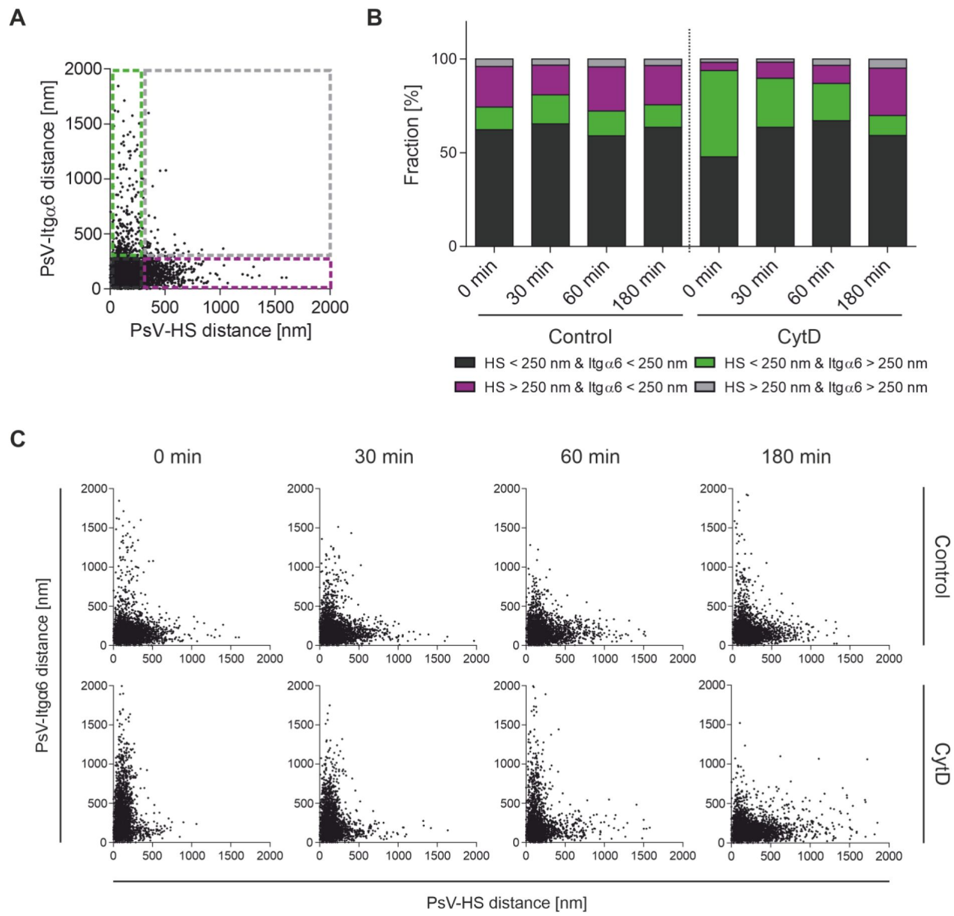


Figure 6.

Pattern of PsV-Itgα6 distances and PsV-HS distances over time.

(A) Definition of four PsV populations. Dashed green rectangle; PsVs with a distance to HS < 250 nm and to Itgα6 > 250 nm. Dashed magenta rectangle; PsVs with a distance to HS > 250 nm and to Itgα6 < 250 nm. Dashed gray square; PsVs with a distance to HS > 250 nm and to Itgα6 > 250 nm. PsVs not included in the previous categories have a distance to HS < 250 nm and to Itgα6 < 250 nm. (B) Based on the experiment shown in **Figure 5**, the PsV fraction size (in percent) of the four populations defined in (A) is illustrated using the same colors as in (A), and using black for PsVs with a distance < 250 nm each to Itgα6 and HS. Shown is the mean of the three biological replicates. For means ± SD and statistical analysis see **Supplementary Table 1**. (C) The same data as in (B), plotting for each PsV the shortest distance to Itgα6 against the shortest distance to HS (pooling the three biological replicates; 3,043 – 4,080 PsVs per plot).

of CytD (**Figure 6C**, lower panels, showing increasing PsV-HS distances). This is the time window where we notice the occurrence of endocytic PsV/CD151 structures (see above and **Supplementary Figure 1**).

Discussion

In this study, we investigate the early stages of HPV16 infection that occur prior to virus endocytosis. Our findings reveal that the dynamics of actin in filopodia, rather than the diffusion of released particles, are crucial for the release of viruses from the adhesive ECM and their subsequent transfer to the cell body. When actin-dependent processes are inhibited, virus particles remain trapped in the ECM, allowing for a more synchronized observation of the infection process. Upon removing the transfer blockade, we observe a rapid translocation of heparan sulfate (HS)-coated virus capsids from the ECM to CD151 assemblies and the cell body. Subsequently, the capsid sheds its HS coat and is endocytosed in association with CD151.

A reversible block of PsV translocation from the ECM to the cell body by CytD

Many viruses use cell surface HS for primary attachment, including herpes simplex virus type 1, human cytomegalovirus, human immunodeficiency virus type 1, adenovirus type 2, dengue virus, hepatitis B virus, and vaccinia virus (Cagno et al., 2019). HPVs could be different as they bind as well to HS of the extracellular basement membrane, at least *in vivo* during the wounding and healing processes required for infection (Day and Schelhaas, 2014; Kines et al., 2009; Schiller et al., 2010).

For studying the translocation of virions from the ECM to the cell surface, we arrest the translocation with the actin-transport inhibitor CytD. After 5 h, we find a 6-fold increase of PsVs in the cell periphery compared to the control (**Figure 2B**). After washing off of CytD, the accumulated PsVs diminish with a half-time of about 15 min. This demonstrates fast translocation of PsVs to the cell surface, further supported by the increase in the PCC between PsVs and CD151 from 0 min to 30 min (**Figure 3C**).

Infectivity is diminished by 30% (**Figure 1E**) when cells are allowed to recover for 19 h after washing off of CytD and PsVs. In comparison to the control, this effect appears moderate, considering that the 5 h incubation time without CytD allows for continuous infection by the PsVs. In addition, slightly harmful effects of the CytD treatment cannot be excluded. The rapid onset of PsV translocation and the moderate effect on infection indicate that CytD only transiently prevents PsV uptake.

It is possible that accumulated PsVs are merely diminished by the wash off, however, the effect of this could not have been very strong, considering the 6-fold accumulation of PsVs after the 5 h incubation with CytD. Moreover, in **Figure 4C**, we measure the diminishment of accumulated PsVs in the cell border region, in this case as percentage of all PsVs. The diminishment in the cell border does not reciprocally reflect an increase of PsVs at the cell body. However, assuming that those PsVs not in the cell border region nor within the boundaries of the cell are not increasing, we conclude that over time the number of PsVs at the cell body increases. Hence, the PsV diminishment at the cell border region is accompanied by an increase of cell body PsVs (provided background PsVs are constant), documenting that PsVs indeed translocate.

Compared to **Figure 2B**, in **Figure 4C** the half-time of diminishment appears longer. We attribute this to an underestimation of the number of PsVs at 0 min/CytD. If the resolution of the accumulated PsVs in **Figure 4C** had been higher, their fraction might have been larger than 36%. Consequently, the half-time of PsV diminishment would be shorter yet. In contrast, the microscope resolution in **Figure 2** is not a limiting factor for the time course measurements, as instead of counting PsVs (**Figure 4C**) we quantify their integrated fluorescence intensity (only possible at the periphery, as autofluorescence precludes quantification of PsVs in the cell body region). Despite of these minor uncertainties, we conclude that the translocation step is fast and efficient.

Some trends in the data suggest that the CytD treatment synchronizes HPV uptake. On the one hand, we see endocytic structures after CytD treatment more frequently (for examples see **Supplementary Figure 1**), in particular between 60 min and 180 min. On the other hand, the CD151 intensity diminishes over time only after the CytD treatment (**Figure 3B**). We speculate that the processing of the PsVs in the ECM promotes desynchronization and is largely completed after the 5 h pre-incubation with CytD. Hence, after CytD removal most PsVs are ready at the same time for cell surface binding. Compared to the control, 6-fold more PsVs (**Figure 2B**) synchronously translocate to the cell surface. Thus, in the following 3 h more endocytic events occur than in the control, which is noticeable in a diminishment of CD151 being co-internalized with the PsVs.

The role of actin-driven virion transport

As outlined above, the electrostatic HPV-HS bond is strong, and thus requires processing of the capsid surface/HS in order to proceed on the infection pathway. This hypothesis is supported by several studies showing that e.g., the activity of HS cleaving enzymes facilitates the virus release from the ECM and infection (Surviladze et al., 2015). CytD supposedly does not inhibit the activity of these enzymes that may be present in the ECM (Jamieson et al., 2021). It is possible that not at every ECM location all enzymes are available, therefore PsVs must be pulled through the ECM to the required enzymes. If that is the case, CytD is indeed not likely to inhibit the processing enzymes, but rather inhibits the transport of PsVs and thus prevents the contact to the necessary enzymes. Alternatively, processing might be completed but virions remain associated with the ECM by electrostatic/van-der-

Waals forces. In this case, for ECM release actin-driven transport would be required to transport the virion out of the sticky matrix to the cell surface. Here, detachment from the ECM may be facilitated by new bonds established between the PsVs and cell surface receptors. Now, the PsVs and cell surface receptors move towards the central cell body by intracellular dynamics and we ultimately observe endocytosis (see **Figure 3A**).

In our experiments, many PsVs reach the cell body after crossing a distance of several micrometers (see the analyzed cell periphery area in **Figure 2A**). It has been previously described that HPV-PsVs are taken up from the ECM via transport along actin-rich protrusions (Schelhaas et al., 2008; Smith et al., 2008). In HeLa cells, the speed of the fastest, straight migrating PsVs is in the range of several micrometer per minute (Schelhaas et al., 2008). Considering that only the fastest PsVs move at that speed, we propose our observed time course is in accordance with the previously reported migration of PsVs along actin-rich protrusions.

If cells require some time to recover from CytD before PsV migration restarts, the translocation half-time of 15 min is likely to be overestimated. In any case, the translocation is fast compared to the entire infection process and therefore cannot largely contribute to the asynchronous HPV uptake.

Inhibition of transport diminishes infection only of subconfluent but not of confluent HaCaT cells by about 50% (Schelhaas et al., 2008). Therefore, actin-dependent translocation along protrusions may be dispensable for infection at a high cell density (Schelhaas et al., 2008) and simply increases the exposure of cells to virions (Smith et al., 2008). However, this increased exposure could be relevant *in vivo*, as wounding of the epidermis results in upregulation of filopodia formation (Vasioukhin et al., 2000). Filipodia usage not only facilitates infection but also increase the likelihood of virions to reach their target cells during wound healing, the filopodia-rich basal dividing cells. In fact, several viruses exploit filopodia during virus entry (Chang et al., 2016), wherefore actin-driven virion transport could have a more important role in HPV infection than currently assumed.

Translocating PsVs are decorated with HS which they lose over time

At 0 min/CytD, 15% of the PsVs are tightly associated with HS (**Figure 5E**), i.e. the distance between PsVs and HS is ≤ 80 nm. 80 nm is above the theoretical distance between an HS fragment adhering to the surface of a PsV and the PsV position. Considering the important role of HS as a primary attachment site, the fraction of 15% appears unexpectedly small. One factor contributing to an underestimation is that accumulated PsVs are not well-resolved (see above). Therefore, some PsV maxima positions actually represent the positions of several merged PsV maxima. Instead of several correct positions that may be within a distance of 80 nm to HS, we obtain a displaced position further away from HS and then these PsVs would not only not contribute to the fraction of tightly associated PsVs, but increase the fraction of non-tightly associated PsVs. This error decreases over time, as the accumulated PsVs diminish and individual PsVs become better resolved. Another reason for underestimation could be that the HS antibody does not reach all HS epitopes in the dense matrix, or that PsV binding and antibody binding to HS is to some extent mutually exclusive. Finally, not all PsVs bound to the ECM are expected to bind to HS but could also bind to laminin 332 (Culp et al., 2006).

The shortening of the HS-Itgα6 distance (**Supplementary Figure 4D**) suggests movement of HS towards the cell body. Moreover, in CytD treated cells, after 30 min, the increase in the PCC between PsVs and HS specifically measured in the cell body region (**Figure 5D**) implies co-migration of PsVs and HS. PsV-HS association decreases in 180 min from 15% to 5% (**Figure 5E**), with 5% being equal to the association in the control (**Figure 5E**). Hence, over time PsVs lose their HS coat. Loss of HS is also visible in **Figure 6C**, showing that over time more PsVs have very long distances to HS. Additionally, the lower HS intensity level in the control (**Figure 5B**) suggests that those PsVs that are actively taken up lose their HS coat. These observations are in line with a previous study showing that the colocalization of HS with cell surface and endocytosed virions decreases over time unless virions are directed into a non-infectious entry pathway (Selinka et al., 2007).

In conclusion, as previously postulated by others (Ozbun and Campos, 2021), our data confirms that PsVs released from the ECM are decorated with HS. However, they lose HS within a few hours after translocation, perhaps due to further structural changes of the capsids or when they bind to receptors for endocytosis.

PsVs-CD151 association is an early contact to the cell surface

CD151 is crucial for cell entry (Scheffer et al., 2013; Spoden et al., 2008). Hence, PsVs must establish contact to CD151 assemblies sometime between the first cell surface contact and endocytosis. As shown in **Figure 3D**, about 10% of the PsVs are tightly associated with CD151 over the entire incubation period if cells are not treated with CytD. We find a similar effect in samples treated with CytD, with the exception of the 0 min/CytD time point, where the value of associated PsVs is almost halved. The reduction could be in some part due to background overcorrection (see legend of **Supplementary Figure 3**), but also mainly because not all PsVs have translocated via actin-driven transport to CD151 receptor assemblies.

In any case, the data suggest that PsVs establish contact to CD151 assemblies at the moment of translocation to the cell surface, and from then on, the fraction of tightly associated PsVs remains on a 10% level. This implies that, after the initial formation of the PsV-CD151 assemblies, they do not change in nature. Yet, we observe agglomerated CD151 maxima with PsVs at later time points that likely are endocytic structures (see arrow in **Figure 3A**, **Supplementary Figure 1**). Therefore, some further CD151 reorganization into larger entry platforms must occur (Florin and Lang, 2018). This is in accordance with the observation that virions enter the cell in crowds with many of them occupying one endocytic organelle (Schelhaas et al., 2012). However, we do not resolve these events in our assay, either because they are too rare or they are not associated with a change in the PsV-CD151 distance.

For the PsVs tightly associated with CD151, one may have expected a larger fraction than 10%. Aside from the issues regarding the PsV-HS association, for the PsV-CD151 association our 80 nm distance criteria may be too strict, because PsVs are not expected to bind directly to CD151 but to larger plasmalemmal assemblies containing, among other proteins, the yet unknown PsV cell surface receptor.

Altogether, we propose that the initial contact between PsVs and the cell surface involves an assembly defined by CD151 (**Figure 7**). Moreover, our data show that after cell surface contact the virions are decorated with HS fragments.

Integration of our data into the cascade of steps underlying HPV infection

HPV infection is the result of several steps, starting with the initial binding of virions via electrostatic and polar interactions (Dasgupta et al., 2011) to the primary attachment site HS, which induces capsid modification (Feng et al., 2024) and HS cleavage (Surviladze et al., 2015), enabling the virion to be released from the ECM or the glycocalyx. Next, virions bind to the cell surface (to a secondary receptor or an early version of it), and become internalized via endocytosis, before they are trafficked to the nucleus (Ozbun and Campos, 2021).

We propose that after 5 h of CytD treatment, glycan-induced structural activation of the capsid and HS cleavage essentially are completed ((i) in **Figure 7**). Subsequently, (ii) the HS-decorated virion is transported from the ECM to the cell body on filopodia, which takes about 15 min. During this process, PsV-CD151 association occurs early. (iii) In the samples recorded 30 – 180 min after CytD wash off, we observe that PsVs lose their HS coat, and individual PsV-CD151 assemblies seem to merge into larger structures that are subsequently endocytosed.

Materials and Methods

Antibodies and PsVs

We used the following primary antibodies in immunostainings. For the capsid protein L1, a rabbit polyclonal antibody (pAb) K75 (1:1000) as described previously (Rommel et al., 2005), for CD151 a mouse monoclonal antibody (mAb) (1:200) (Biorad, cat# MCA1856GA), for Heparan sulfate (HS) a mouse IgM monoclonal antibody (1:200) (amsbio, cat# 370255-S) and for Itgα6 a rabbit polyclonal antibody (1:200) (Invitrogen™, cat# PA5-12334). As secondary antibodies we used a STAR GREEN-coupled goat-anti-rabbit (1:200) (Abberior, cat# STGREEN-1002-500UG), an AlexaFluor™ 594-coupled donkey-anti-rabbit (1:200) (Invitrogen™, cat# A21207), an AlexaFluor™ 594-coupled donkey- anti-mouse (1:200) (Invitrogen™, cat# A21203), an AlexaFluor™ 594-coupled donkey-anti-mouse IgM (1:200) (Invitrogen™, cat# A21044), a STAR RED-coupled goat-anti-mouse (1:200) (Abberior, cat# STRED-1001-500UG) and a STAR RED-coupled goat-anti-rabbit (1:200) (Abberior, cat# STRED-1002- 500UG). Additionally, phalloidin iFluor488 (1:1000 of ready to use solution) (abcam, cat# ab176753) or phalloidin iFluor647 (1:1000) (abcam, cat# ab176759) was used to stain

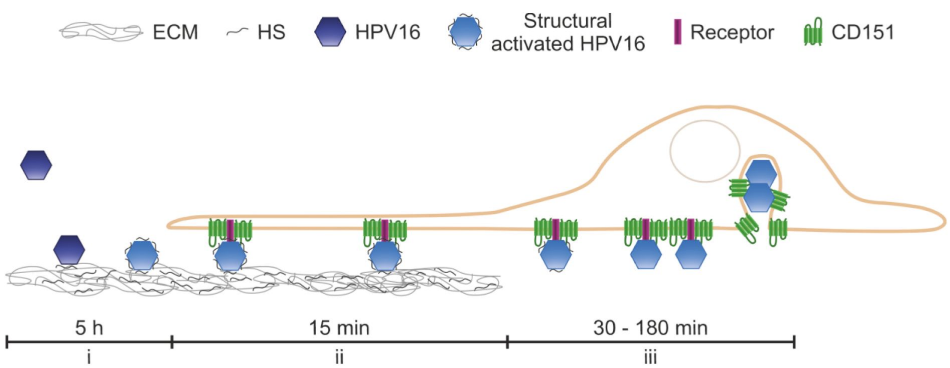


Figure 7.

Model figure.

The structural activation of PsVs in the ECM is completed within 5 h (i). Within 15 min, the structurally- activated HPV16 translocate from the ECM to CD151 assemblies, likely located at the filopodia, and move toward the cell body (ii). During this process, the virions retain some heparan sulfate (HS) on their surface. (iii) Eventually, they lose their HS coat, and individual PsV-CD151 assemblies merge into larger complexes, which are subsequently endocytosed.

F-actin. Staining of the PsV plasmid DNA was performed by click-labeling according to the manufacturer's instructions with the dye 6-FAM Azide (Baseclick EdU 488 kit, Carl Roth, cat# 1Y67.1).

HPV16 PsVs were produced following established procedures (Buck et al., 2004; Spoden et al., 2012). HEK293TT cells were cultured in 175 cm² flasks and transfected with polyethylenimine (PEI), using equimolar amounts of codon-optimized HPV16 L1/L2 (pShell 16L1/L2wt) and a reporter plasmid (pGL4.20-puro-HPV16 LCR). For the production of 5-ethynyl-2'-deoxyuridine (EdU)-labeled PsVs, the culture medium was replaced 5 hours post-transfection with fresh medium containing 30 µM EdU. After 48 hours, cells were harvested, centrifuged, and washed twice with phosphate-buffered saline (PBS) supplemented with 9.5 mM $MgCl_2$ (1x PBS/ $MgCl_2$). Following a final centrifugation step, the pellet was resuspended in 1x PBS/ $MgCl_2$ containing 0.5% Brij58 (Sigma-Aldrich), and 0.2% Benzonase (Merck Millipore), and incubated for 24 hours at 37°C on a rotating platform. Subsequently, lysates were chilled on ice and the NaCl concentration was adjusted to 0.85 M. After clarification by centrifugation, the supernatant was loaded onto an iodixanol (Optiprep) gradient consisting of 39%, 33%, and 27% layers (bottom to top). Gradients were equilibrated for 90 minutes at room temperature before ultracentrifugation at 55,000 rpm for 3.5 hours at 16°C. Fifteen fractions of 300 µl each were collected from the top and analyzed via luciferase reporter assay to identify peak fractions. PsV titers were determined based on packaged genomes (viral genome equivalents, vge) as previously described (Spoden et al., 2012). The concentrations of stock solutions are 7,7 x 10^6 vge/µl (for microscopy experiments) and 14,1 x 10^6 vge/µl (for the luciferase assay).

Cell culture

For microscopy, human immortalized keratinocytes (HaCaTs) (Cell Lines Services, Eppelheim, Germany) are cultured in high glucose DMEM (Gibco®, cat# 61965-026) medium supplemented with 10% FBS (PAN Biotech, cat# P30-3031) and 1% Penicillin/Streptomycin (10,000 U/ml Penicillin, 10 mg/ml Streptomycin; PAN Biotech, cat# P06-07100) at 37 °C with 5 % CO_2. For the luciferase infection assay, cells were grown at 37°C in Dulbecco's modified Eagle's medium (DMEM + GlutaMAX (Thermo Fisher Scientific), supplemented with 10% FBS (Sigma-Aldrich), 1% minimum essential medium non-essential amino acids (MEM non-essential amino acids (Thermo Fisher Scientific)), and 5 µg/ml ciprofloxacin (Fresenius Kabi). For experiments, the antibiotics in the medium are omitted.

Sample preparation and immunostaining

About 150,000 HaCaT cells were plated onto 25 mm diameter poly-L-lysine (PLL) coated glass coverslips in 6-well plates and incubated for 24 h at 37°C and 5% CO_2. The next day, cells were incubated at 37°C and 5% CO_2 for 5 h with PsVs (50 vge/plated cell) and 10 µg/ml cytochalasin D (CytD; stock solution 10 mg/ml in dimethyl sulfoxide (DMSO)) (Life Technologies, cat# PHZ1063) in DMEM supplemented with 10% FBS. For controls, we added the same amount of DMSO without CytD. Cells were washed with PBS (137 mM NaCl, 2.7 mM KCl, 1.76 mM KH_2PO_4, 10 mM Na_2HPO_4, pH7.4) and fresh DMEM supplemented with 10% FBS was added. Cells were incubated further at 37°C and 5% CO_2 for 0 min (here the medium was added followed by immediate removal), 15 min, 30 min, 60 min and 180 min. Cells were washed twice with PBS and fixed at room temperature (RT) with 4% PFA in PBS for 30 min, unless membrane sheets are generated. In this case, the coverslips were placed in ice cold sonication buffer (120 mM KGlu, 20 mM KAc, 10 mM EGTA, 20 mM HEPES, pH 7.2) and a 100 ms ultrasound pulse at 100% power was applied. This was repeated until in total about 10 pulses are applied at different locations of the coverslip. Then, membrane sheets were fixed like cells at RT with 4% PFA in PBS for 30 min. PFA is removed, and residual PFA is quenched by 50 mM NH_4Cl in PBS for 30 min. Afterwards, samples are blocked with 3% BSA in PBS for 30 min. Staining of PsVs was performed by click-labeling with the dye 6-FAM for 30 min at RT according to the manufacturer's instructions. Then, samples were washed three times with PBS. In case of no PsV labeling by click-chemistry, directly after blocking, the respective primary antibodies are added: mouse IgM mAb against HS (1:200), rabbit pAb against Itgα6 (1:200), mouse mAb against CD151 (1:200) and rabbit pAb K75 against L1 (1:1000) in 3% BSA in PBS for 2 h. Samples were washed three times with PBS before adding the respective secondary antibodies and fluorescent labelled phalloidins in 3% BSA in PBS for 1 h: for HS we add AlexaFluor™ 594 coupled to donkey-anti-mouse IgM (1:200), for Itgα6 STAR RED coupled to goat-anti-rabbit (1:200), for CD151 AlexaFluor™ 594 coupled to donkey-anti-mouse (1:200) or STAR RED coupled to goat-anti-mouse (1:200), for L1 AlexaFluor™ 594 coupled to donkey-anti-rabbit (1:200) or STAR GREEN coupled to goat-anti-rabbit (1:200) and for F-actin phalloidin iFluor488 (1:1000) or phalloidin iFluor647 (1:1000). Afterwards, samples are washed three times with PBS, followed by mounting of the coverslips onto microscopy slides using ProLong® Gold antifade mounting medium (Invitrogen, cat# P36930).

Confocal and STED microscopy

For confocal and STED microscopy, samples were imaged employing a 4-channel STED microscope from Abberior Instruments (available at the LIMES institute imaging facility, Bonn, Germany). The microscope is based on an Olympus IX83 confocal microscope equipped with an UPlanSApo 100x (1.4 NA) objective (Olympus, Tokyo, Japan). Confocal and STED micrographs are recorded simultaneously. For confocal imaging, a 485 nm laser is used for the excitation of 6-FAM, STAR GREEN and phalloidin iFluor488, and emission is detected at 500–550 nm. For STED imaging, a 561 nm laser (at 45%) is used for the excitation of AlexaFluor™ 594 (detection at 580–630 nm) and a 640 nm laser (at 45%) for the excitation of STAR RED and phalloidin iFluor647 (detection at 650-720 nm), in combination with a 775 nm laser for depletion (at 45%). The pixel size is set to 25 nm.

Analysis of confocal and STED micrographs

Images are analyzed using Fiji ImageJ in combination with a custom written macro, essentially as described previously (Schmidt et al., 2024). When analyzing double stainings of Alexa594 (red channel) and STAR RED (long red channel), we corrected for crosstalk from the red into the long red channel by subtracting 40% of the intensity of the red channel image from the long-red channel image. First, images are smoothed with a Gaussian blur to improve maxima detection. For confocal PsV-DNA micrographs, we employed a Gaussian blur of $\sigma = 1$, and for confocal PsV-L1 and STED micrographs a Gaussian blur of $\sigma = 0.5$. Local maxima were detected using the 'Find Maxima' function (noise tolerance 60 for L1 if stained together with phalloidin iFluor488, and 80 if stained together with CD151, 3 for click-labeled PsVs, 8 for CD151, 6 for HS and 15 for Itgα6), yielding maxima positions in pixel positions. For analysis, rectangular regions of interest (ROIs) are defined. To measure the L1 maxima intensity (**Figure 1C** and **D**) the ROIs covered the whole image. For CytD wash off experiments the rectangular ROIs covered mainly the cell body (75%) but included the cell periphery as well (25%).

In order to define the cell border region (**Figure 4B**), the ImageJ 'Make Binary' function is used on CD151 STED micrographs for generating a binary mask. When possible, the 'Wand' tool is used to further outline the area covered by the cell, otherwise the same was done manually using the original micrograph as a reference. From the outline in the binary mask, a ROI is created, that is then filled in white with the 'Flood Fill' Tool. The ROI is shrunk by 20 pixels, and cleared inside, what leaves a 20 pixels broad, closed ribbon marking (i) the intracellular side of the cell and (ii) arbitrary edges produced in the process above. These edges are removed by manual adjustments with the 'Pencil' tool and the clear function. Afterwards, the remaining border region defines the intracellular side of the cell border. From this border region a ROI is created by applying the 'Wand' tool, that is symmetrically broadened to 40 pixels. From the 40 pixels, 30 pixels cover the intracellular and 10 pixels the extracellular side. Using the ImageJ 'Find Maxima' function, PsV-L1 maxima are detected (at a noise tolerance of 80) both in the cell border region ROI and in a ROI covering the entire micrograph. From these values, the percentage of PsVs within the cell border region is calculated.

For measuring e.g. the CD151 intensity over time, we used ROIs as illustrated in **Supplementary Figure 2A**, and measured in the ROI the mean gray value. For background correction, we subtracted the mean gray value measured in a ROI next to the cell.

For measuring maxima intensity, a ROI with a diameter of 125 nm (5 pixels) was placed onto the determined maxima positions (see above). Using these ROIs, the average mean gray value of each maximum is measured. The average background mean gray value is measured in a ROI placed next to the cell and subtracted from the average mean gray value of the maxima.

To measure the shortest distance, e.g. between PsV and CD151 maxima, as a quality control for the PsV maxima (CD151 maxima do not undergo the quality control step) we placed onto each maximum position a horizontal and a vertical linescan (31 pixels long × 3 pixels width). Afterwards, we fitted a Gaussian distribution to the intensity distribution of each maximum. Only if at least one of the fits exhibits a fit quality of $R^2 > 0.8$ and if the Gaussian maximum locates central to the intensity distribution the maximum is included in the further analysis. Moreover, using a 125 nm ROI at the PsV and CD151 maxima positions, the center of mass of fluorescence is determined, yielding the maxima positions in sub-pixels. Based on these positions we further calculated the shortest distance of a PsV maximum to the nearest CD151 maximum.

Using the shortest distances, we either calculated the fraction of tightly associated PsVs over time (the fraction of PsVs with a distance ≤ 80 nm to e.g. CD151), or the average shortest distance over time. The fraction of tightly associated PsVs is further corrected for random background association. Random background association increases with the maxima density. The relationship between background association and maxima density is expressed in a linear calibration line. For the calibration line, we use the same images and determine the fraction of PsVs with a distance to e.g. CD151 measured on horizontally and vertically flipped (randomized) ROI-defined images (see also **Supplementary Figure 2A**). The fraction is plotted against the maxima density. Fitting of a linear regression line yields the mathematical relationship between the fraction of tightly associated PsVs and maxima density, from which we obtain the value to be subtracted for background correction.

The Pearson correlation coefficient (PCC) was calculated with a custom written ImageJ macro in ROIs as illustrated in e.g. **Supplementary Figure 2A**. As a control we performed the analysis on horizontally and vertically flipped ROI-defined images as well (see e.g. **Supplementary Figure 2A**). The cell body PCC between PsVs and HS is measured in smaller ROIs covering exclusively the cell body region (this is confirmed with the Itgα6 image as reference).

Per condition and biological replicate, 14 - 15 images are analyzed. For images from one set of experiments, in the figure the same channels are shown at the same scaling and same lookup-tables.

Epi-fluorescence microscopy and image analysis

For epifluorescence microscopy, we used microscopic equipment and settings as previously described (Homsi et al., 2014) except of the illumination system, which is replaced by a Lumencor Light engine® Spectra 90-10034 (Lumencor, USA). In brief, PFA-fixed cells are imaged in PBS containing TMA-DPH [1-(4-tri-methyl-ammonium-phenyl)-6-phenyl-1,3,5-hexatriene-p-toluenesulfonate (T204; Thermofisher)] for visualizing the membranes in the blue channel. PsVs, HS and Itgα6 were imaged in the green, red and far-red channels, respectively.

For analysis, rectangular ROIs of ≈ 45 x 10^3 pixel are placed onto the cell body such that one half of the ROI covers roughly the ECM area and the other one the cell body. The cell periphery is defined using the TMA-DPH image from which a binary mask is created. With reference to the binary mask, a band of 30 pixels width is created beginning at the cell body and reaching out towards the cell periphery. Using this ROI, in the PsV-DNA image the size of the ROI and the mean grey values are measured from which the integrated intensities are calculated after background correction.

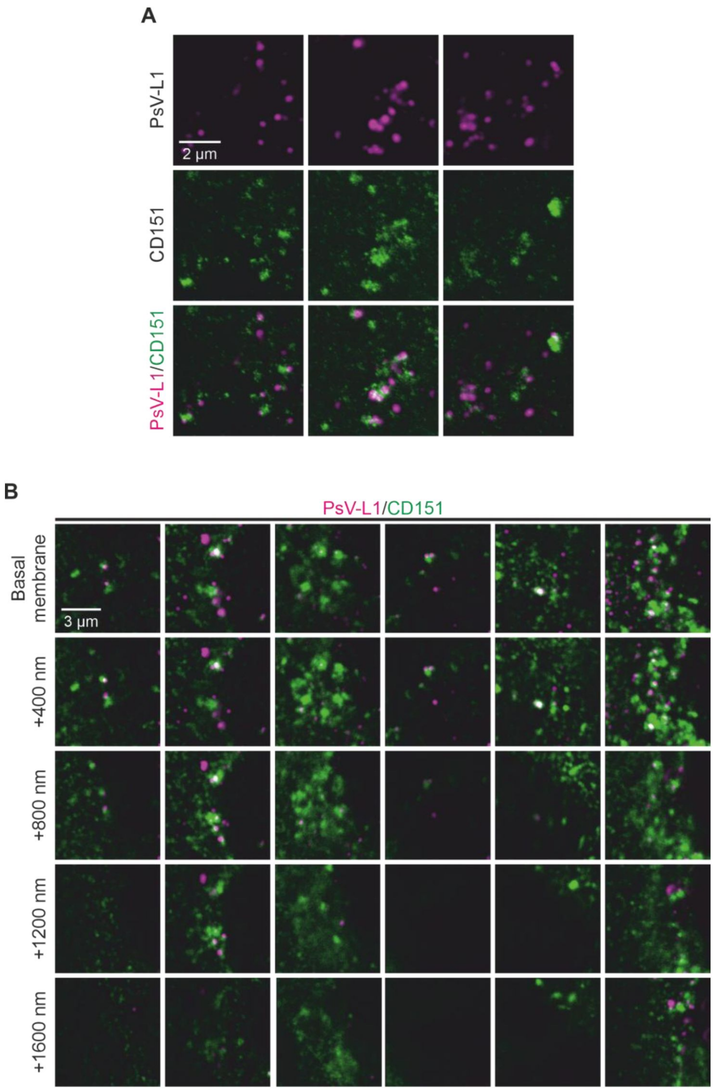
Luciferase infection assay

HaCaT cells are grown in 24-well plates and allowed to adhere overnight. The following day, cells are treated with HPV16 PsVs at a concentration of 100 vge per cell, in the presence or absence of 10 µg/ml cytochalasin D (CytD) in DMSO or an equivalent amount of DMSO (control). In one condition, PsVs/CytD are applied for 5 hours, after which the medium is replaced with fresh medium lacking the compounds, and incubation is continued for additional 19 hours (in total 24 h). In another condition, cells are exposed to PsVs/CytD continuously for the full 24 h period. Then, cells are washed with PBS and lysed using 1 x Cell Culture Lysis Reagent (Promega, cat# E153A). Following high-speed centrifugation, luciferase activity in the cleared lysates is quantified using an LB 942 Tristar 3 luminometer (Berthold Technologies). Cytotoxic effects, accompanied by a loss of membrane integrity (indicated by released lactate dehydrogenase (LDH)), are determined by measuring the LDH levels in the cell lysates. LDH activity is assessed using the CytoTox-ONE™ Homogeneous Membrane Integrity Assay (Promega, cat# G7891). LDH fluorescence is quantified according to the manufacturer’s instructions using the LB 942 Tristar 3 luminometer.

Statistics

Data sets were based on three biological replicates, including 14 - 15 images per replicate. For infection assay, data sets were based on three biological replicates, with three technical replicates from each biological replicate. Data was tested for significance with a two-tailed, unpaired student’s *t* test with significance * = $p < 0.05$, ** = $p < 0.01$ and *** = $p < 0.001$.

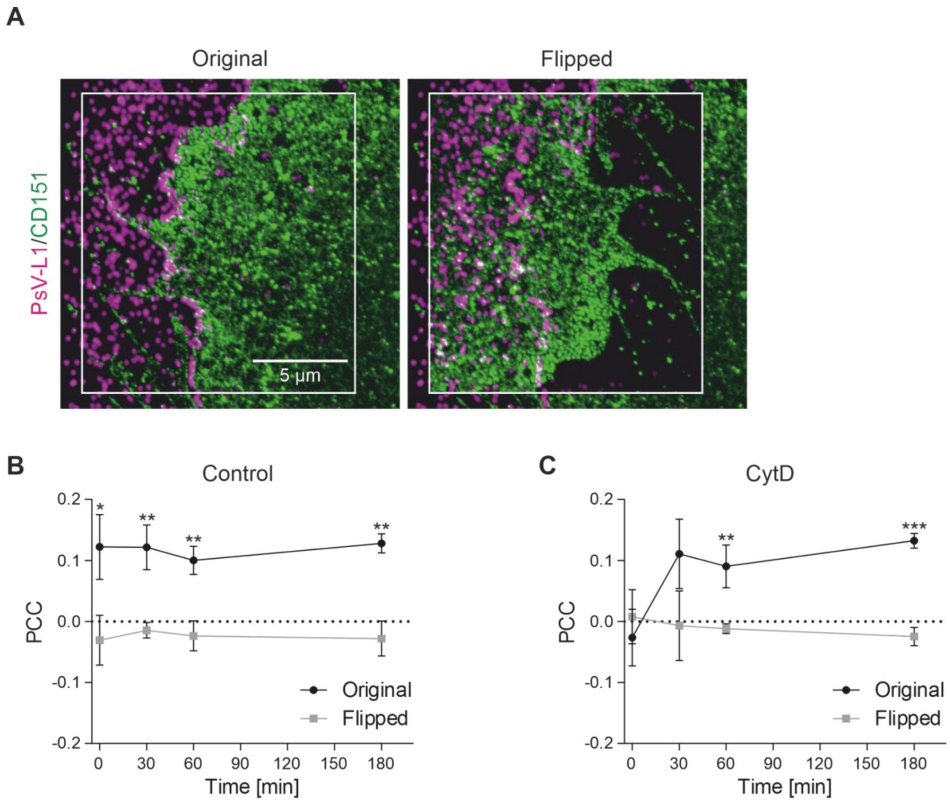
Supplementary information



Supplementary Figure 1.

Examples of agglomerated CD151 maxima that are associated with PsVs and presumably are endocytic structures.

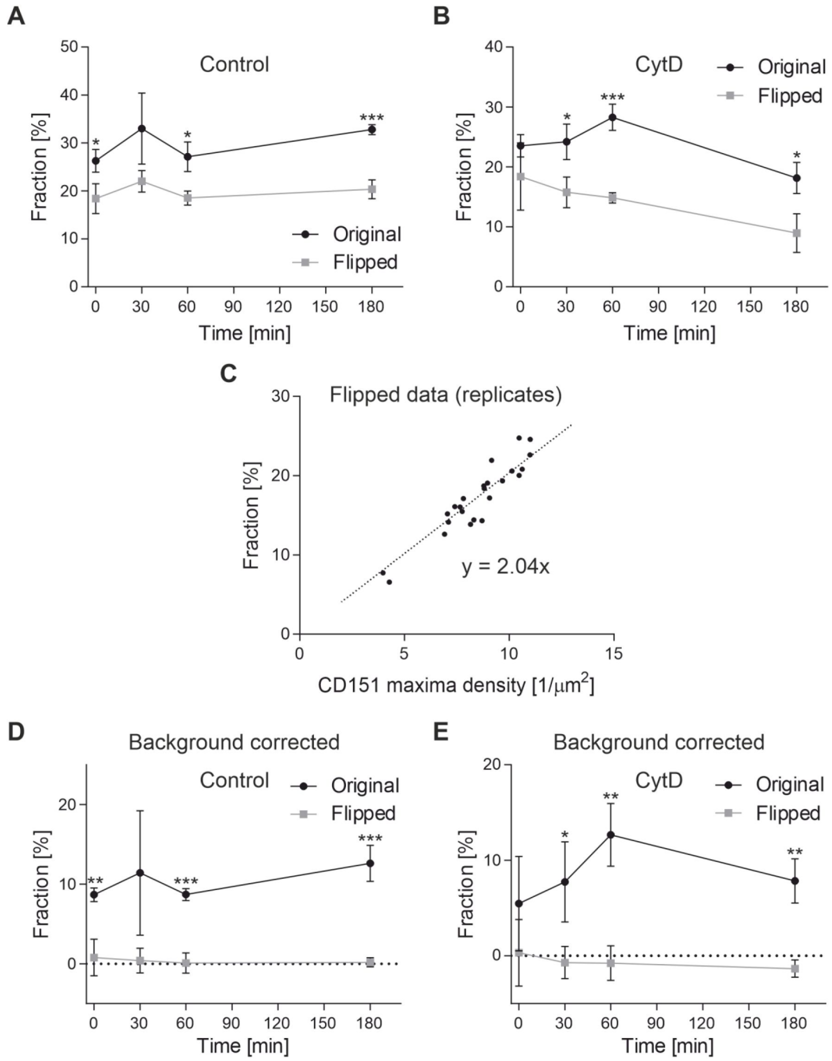
(A) From the data described in **Figure 3** (180 min/CytD), we show more examples of agglomerated CD151 maxima (see green patches) that associate with PsVs (magenta) and likely are endocytic structures. (B) The same experiment as in (A), but cells are scanned with 400 nm steps in the axial direction at confocal resolution, starting at the basal membrane. The structures noticed in the basal membrane as agglomerated CD151 maxima continue deeper into the cell. In some cases, more than a micrometer (see second example from the left).



Supplementary Figure 2.

PCC controls.

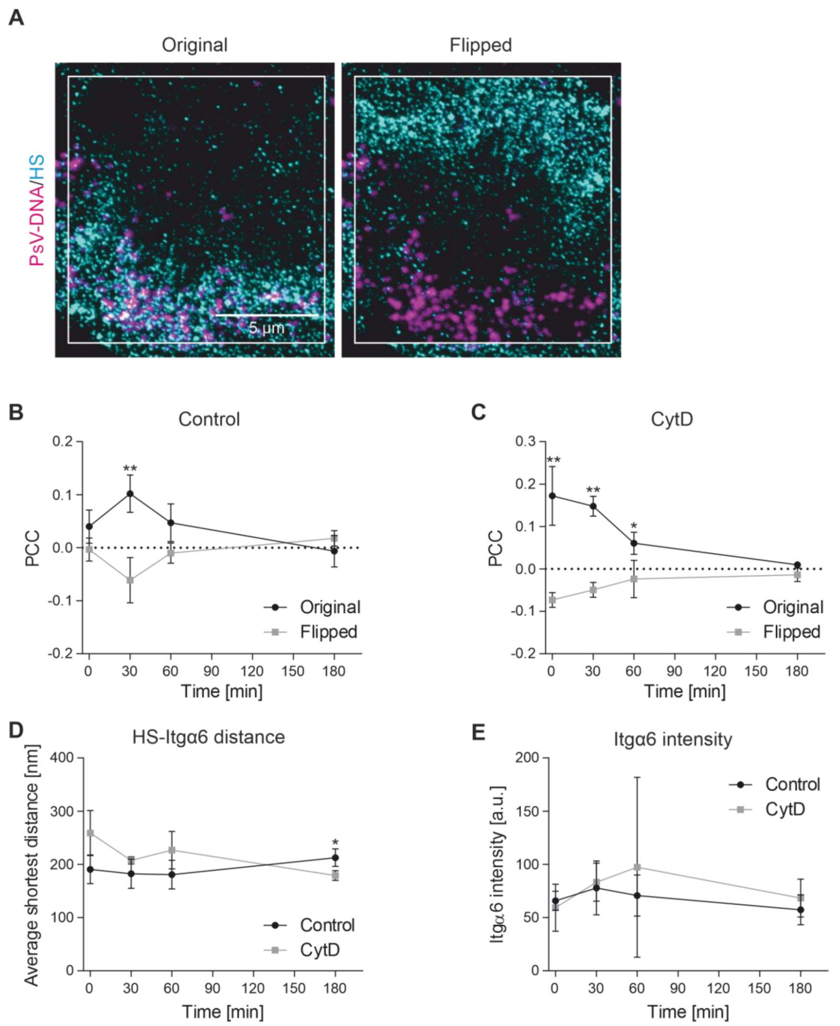
(A) In order to determine the background colocalization, we analyze the PCC on flipped images as well. Left, original image taken from the 0 min/CytD condition. The ROI selected for analysis is shown as white box. Right, the green image within the ROI is flipped horizontally and vertically. Now, a pair of images with the same density of objects and intensity distributions can be analyzed. (B) PCC between PsV-L1 and CD151 of the control condition (**Figure 3C**) is shown again, together with the respective PCCs determined on flipped images. (C) The PCC between PsV-L1 and CD151 of the CytD condition (**Figure 3C**) is shown again, together with the respective PCCs determined on flipped images. Statistical difference between the same time points of original and flipped images is analyzed by using the two-tailed, unpaired student's *t* test (n = 3). Values are given as mean ± SD.



Supplementary Figure 3.

Background correction of the fraction of PsVs tightly associated with CD151.

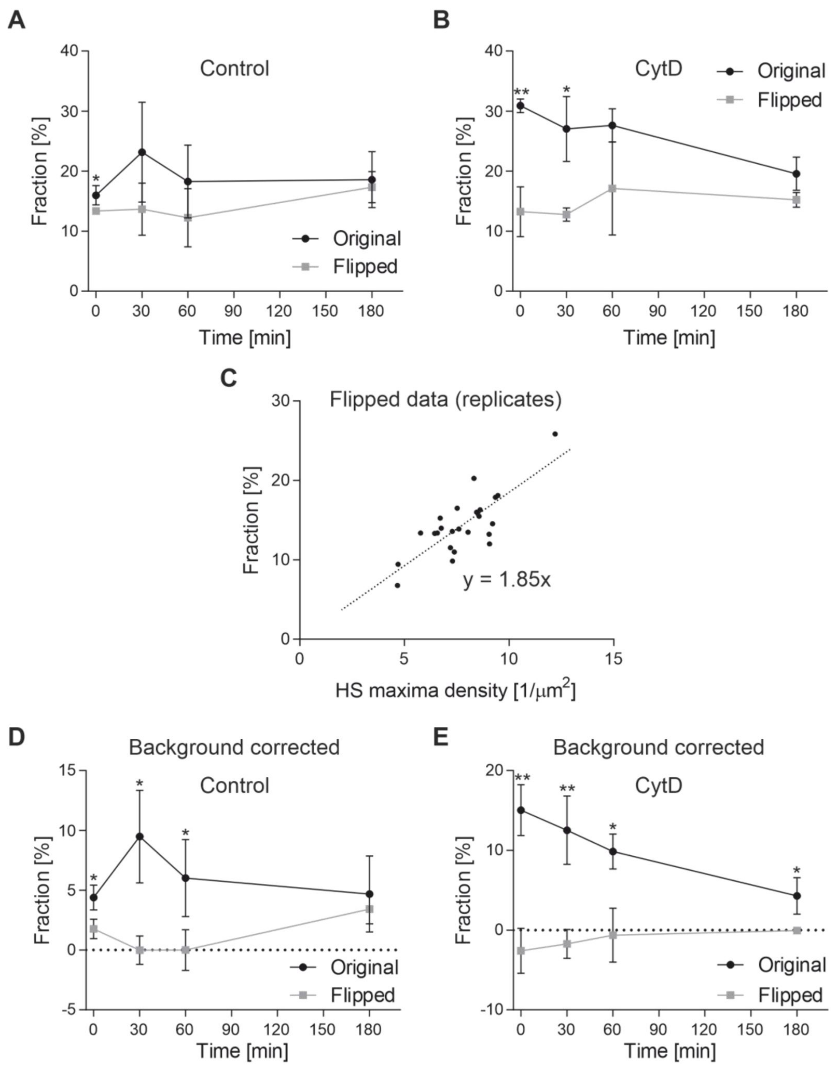
(A) The fraction of tightly associated PsVs (PsV-L1 maxima with a distance ≤ 80 nm to the next nearest CD151 maximum) in the control of the experiment described in **Figure 3**, analyzed on original and flipped images (for an example of a flipped image see **Supplementary Figure 2A**). On flipped images, the fractions often are more than half of the original images. This demonstrates that many of the PsVs have a distance ≤ 80 nm to CD151 merely by chance (random background association). (B) Same as (A) for the CytD condition. (C) Each flipped data point in (A) and (B) is the average of three biological replicates. From the altogether 24 replicates of the flipped images, we plot the fraction of tightly associated PsVs against the CD151 maxima density. As expected, the fraction increases with the maxima density. The fitted linear regression line describes how the background association depends on the maxima density. With the equation of the regression line the random background association can be calculated, for any maxima density in an original images the random background association can be calculated. (D) For each replicate from the original and the flipped images in (A), the background fraction is calculated using the equation of the regression line in (C) and the respective CD151 maxima density, and subtracted from the fraction of tightly associated PsVs. (E) Same as (D) for the CytD condition. The corrected original values in (D) and (E) are shown in **Figure 3D**. Please note that the data point CytD/0 min likely is overcorrected because we flip a CD151 rich region onto the accumulated PsVs that in the original image are in a region that is low in CD151. Statistical difference between the same time points of original and flipped images is analyzed by using the two-tailed, unpaired student's *t* test (n = 3). Values are given as mean ± SD.



Supplementary Figure 4.

PCC controls, HS-Itgα6 average shortest distance and Itgα6 intensity.

Additional analysis from the experiment shown in **Figure 5**. (A) Example of flipped HS image used as control (for explanation please see legend of **Supplementary Figure 2**). Left, original image taken from the 0 min/CytD condition. The ROI selected for analysis is shown as white box. Right, the cyan image within the ROI is flipped horizontally and vertically. (B) The PCC between PsV-DNA and HS of the control condition (**Figure 5C**) is shown again, together with the respective PCCs determined on flipped images. (C) The PCC between PsV-DNA and HS of the CytD condition (**Figure 5C**) is shown again, together with the respective PCCs determined on flipped images. (D) The average shortest distance of HS maxima to the next nearest Itgα6 maximum and (F) the average Itgα6 intensity. Statistical difference between the same time points of original and flipped images or control and CytD is analyzed by using the two-tailed, unpaired student's *t* test (n = 3). Values are given as mean ± SD.



Supplementary Figure 5.

Background correction of the fraction of PsVs tightly associated with HS.

(A) The fraction of tightly associated PsVs (PsV-DNA maxima with a distance ≤ 80 nm to the next nearest HS maximum) in the control of the experiment described in **Figure 5**, analyzed on original and flipped images (for an example of a flipped image see **Supplementary Figure 4A**). (B) Same as (A) for the CytD condition. (C) From the altogether 24 replicates of the flipped images, the fraction of tightly associated PsVs is plotted against the HS maxima density, and a linear regression line is fitted to the data points. (D) For each replicate from the original and the flipped images in (A), the background fraction is calculated using the equation of the regression line in (C) and the respective HS maxima density, and subtracted from the fraction of tightly associated PsVs. (E) Same as (D) for the CytD condition. The background corrected original values in (D) and (E) are shown in **Figure 5E**. Statistical difference between the same time points of original and flipped images is analyzed using the two-tailed, unpaired student's *t* test (n = 3). Values are given as mean ± SD.

Fraction of PsVs [%]	0 min			30 min			60 min			180 min		
	control	CytD	p-value	control	CytD	p-value	control	CytD	p-value	control	CytD	p-value
HS < 250 nm & Itga6 < 250 nm	62.31 ± 5.05	47.83 ±3.26	**0.0271**	65.48 ± 8.35	63.67 ± 1.14	0.7764	59.06 ± 10.64	67.16 ± 9.84	0.4739	63.69 ± 4.27	59.28 ± 3.17	0.3065
HS < 250 nm & Itga6 > 250 nm	12.18 ± 5.51	46.04 ± 3.67	**0.0019**	15.46 ± 3.14	26.12 ± 2.16	**0.0167**	13.25 ± 1.94	19.87 ± 3.12	0.0637	11.97 ± 1.53	10.64 ± 1.15	0.3808
HS > 250 nm & Itga6 < 250 nm	21.59 ± 4.07	4.34 ± 0.37	**0.0040**	15.83 ± 5.73	8.58 ± 2.33	0.1723	23.49 ± 6.55	9.60 ± 4.81	0.0731	20.88 ± 3.31	25.24 ± 1.96	0.1842
HS > 250 nm & Itga6 > 250 nm	3.93 ± 0.30	1.79 ± 0.60	**0.0110**	3.23 ± 0.88	1.63 ± 0.21	0.0675	4.19 ± 2.43	3.38 ± 2.20	0.7425	3.46 ± 0.54	4.84 ± 2.87	0.5393

Supplementary Table 1.

Means ± SD of the data shown in Figure 6B.

P-values between control and CytD are calculated by using the two-tailed, unpaired student's *t* test (n = 3). Significant p-values are illustrated in bold.

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Reviewer #1 (Public review):

The authors' goal was to arrest PsV capsids on the extracellular matrix using cytochalasin D. The cohort was then released, and interaction with the cell surface, specifically with CD151, was assessed.

The model that fragmented HS associated with released virions mediates the dominant mechanism of infectious entry has only been suggested by research from a single laboratory and has not been verified in the 10+ years since publication. The authors are basing this study on the assumption that this model is correct, and these data are referred to repeatedly as the accepted model despite much evidence to the contrary. The discussion in lines 65-71 concerning virion and HSPG affinity changes is greatly simplified. The structural changes in the capsid induced by HS interaction and the role of this priming for KLK8 and furin cleavage have been well researched. Multiple laboratories have independently documented this. If this study aims to verify the shedding model, additional data need to be provided. The model should be fitted into established entry events, or at minimum, these conflicting data, a subset of which is noted below, need to be acknowledged.

(1) The Sapp lab (Richards et al., 2013) found that HSPG-mediated conformational changes in L1 and L2 allowed the release of the virus from primary binding and allowing secondary receptor engagements in the absence of HS shedding.

(2) Becker et al. found that furin-precleaved capsids could infect cells independently of HSPG interaction, but this infection was still inhibited with cytochalasin D.

(3) Other work from the Schelhaas lab showed that cytochalasin D inhibition of infection resulted in the accumulation of capsids in deep invaginations from the cell surface, not on the ECM.

(4) Selinka et al., 2007, showed that preventing HSPG-induced conformational changes in the capsid surface resulted in noninfectious uptake that was not prevented with cytochalasin D.

(5) The well-described capsid processing events by KLK8 and furin need to be mechanistically linked to the proposed model. Does inhibition of either of these cleavages prevent engagement with CD151?

The authors need to consider an explanation for these discrepancies.

Other issues:

(1) Line 110-111. The statement about PsVs in the ECM being too far away from the cell surface to make physical contact with the cell surface entry receptors is confusing. ECM binding has not been shown to be an obligatory step for in vitro infection. This idea is referred to again on lines 158-159 and 199. The claim (line 158) that PsV does not interact with the cell within an hour needs to be demonstrated experimentally and seems at odds with multiple laboratories' data. PsV has been shown to directly interact with HSPG on the cell surface in addition to the ECM. Why are these PsVs not detected?

(2) The experiments shown in Figure 5 need to be better controlled. Why is there no HS staining of the cell surface at the early timepoints? This antibody has been shown to recognize N-sulfated glucosamine residues on HS and, therefore, detects HSPG on the ECM and cell surface. Therefore, the conclusion that this confirms HS coating of PsV during release from the ECM (line 430-431) is unfounded. How do the authors distinguish between "HS-coated virions" and HSPG-associated virions?

It is difficult to comprehend how the addition of 50 vge/cell of PsV could cause such a global change in HS levels. The claim that the HS levels are decreased in the non-cytochalasin-treated cells due to PsV-induced shedding needs to be demonstrated. If HS is actually shed, staining of the cell periphery could increase with the antibody 3G10, which detects the HS neoepitope created following heparinase cleavage.

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Reviewer #2 (Public review):

Summary:

Massenberg and colleagues aimed to understand how Human papillomavirus particles that bind to the extracellular matrix (ECM) transfer to the cell body for later uptake, entry, and infection. The binding to ECM is key for getting close to the virus's host cell (basal keratinocytes) after a wounding scenario for later infection in a mouse vaginal challenge model, indicating that this is an important question in the field.

Strengths:

The authors take on a conceptually interesting and potentially very important question to understand how initial infection occurs in vivo. The authors confirm previous work that actin-based processes contribute to virus transport to the cell body. The superresolution microscopy methods and data collection are state-of-the art and provide an interesting new way of analysing the interaction with host cell proteins on the cell surface in certain infection scenarios. The proposed hypothesis is interesting and, if substantiated, could significantly advance the field.

Weaknesses:

As a study design, the authors use infection of HaCaT keratinocytes, and follow virus localisation with and without inhibition of actin polymerisation by cytochalasin D (cytoD) to analyse transfer of virions from the ECM to the cell by filopodial structures using important cellular proteins for cell entry as markers.

First, the data is mostly descriptive besides the use of cytoD, and does not test the main claim of their model, in which virions that are still bound to heparan sulfate proteoglycans are transferred by binding to tetraspanins along filopodia to the cell body.

Second, using cytoD is a rather broad treatment that not only affects actin retrograde flow, but also virus endocytosis and further vesicular transport in cells, including exocytosis. Inhibition of myosin II, e.g., by blebbistatin, would have been a better choice as it, for instance, does not interfere with endocytosis of the virus.

Third, the authors aim to study transfer from ECM to the cell body and the effects thereof. However, there are substantial, if not the majority of, viruses that bind to the cell body compared to ECM-bound viruses in close vicinity to the cells. This is in part obscured by the small subcellular regions of interest that are imaged by STED microscopy, or by the use of plasma membrane sheets. As a consequence, the obtained data from time point experiments is skewed, and remains for the most part unconvincing due to the fact that the origin of virions in time and space cannot be taken into account. This is particularly important when interpreting association with HS, the tetraspanin CD151, and integral alpha 6, as the low degree of association could originate from cell-bound and ECM-transferred virions alike.

Fourth, the use of fixed images in a time course series also does not allow for understanding the issue of a potential contribution of cell membrane retraction upon cytoD treatment due to destabilisation of cortical actin. Or, of cell spreading upon cytoD washout. The microscopic analysis uses an extension of a plasma membrane stain as a marker for ECM-bound virions, which may introduce a bias and skew the analysis.

Fifth, while the use of randomisation during image analysis is highly recommended to establish significance (flipping), it should be done using only ROIs that have a similar density of objects for which correlations are being established. For instance, if one flips an image with half of the image showing the cell body, and half of the image ECM, it is clear that association with cell membrane structures will only be significant in the original. I am rather convinced that using randomisation only on the plasma membrane ROIs will not establish any clear significance of the correlating signals. Also, there should be a higher n for the measurements.

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Author response:

> ***Reviewer #1 (Public review):***
>
> *The authors' goal was to arrest PsV capsids on the extracellular matrix using cytochalasin D. The cohort was then released, and interaction with the cell surface, specifically with CD151, was assessed.*
>
> *The model that fragmented HS associated with released virions mediates the dominant mechanism of infectious entry has only been suggested by research from a single laboratory and has not been verified in the 10+ years since publication. The authors are basing this study on the assumption that this model is correct, and these data are referred to repeatedly as the accepted model despite much evidence to the contrary.*

Please note that we state in the introduction on line 65/66 Two release mechanisms are discussed, that mutually are not exclusive.´ This is implying that we do not consider the shedding model as the one accepted model. HS may associate with PsVs despite of a decreased affinity and only after priming (see below the ‘priming model’) may translocate to the cell body.

Furthermore, we do not state in the discussion either that the shedding model is the preferred one; although it is correct that we refer to the shedding model more extensively, simply because we find HS associated with transferred PsVs, which is in line with this model and requires its citation.

> *The discussion in lines 65-71 concerning virion and HSPG affinity changes is greatly simplified. The structural changes in the capsid induced by HS interaction and the role of this priming for KLK8 and furin cleavage have been well researched. Multiple laboratories have independently documented this. If this study aims to verify the shedding model, additional data need to be provided.*

As outlined above, our finding is compatible with both models, and we do not aim to verify the shedding model or disprove the priming model.

It appears that the referee wishes more visibility of the priming model. Inhibition of KLK8 and furin should reduce the translocation to the cell body, no matter whether PsVs carry HS on their surface or not. For revision, we plan an experiment as in Figure 3 (CytD), testing whether either KLK8 or furin inhibition blocks the transfer to the cell body. Then, our data can be discussed also in the context of the priming model and by this increase its visibility.

> *The model should be fitted into established entry events, or at minimum, these conflicting data, a subset of which is noted below, need to be acknowledged.*
>
> *(1) The Sapp lab (Richards et al., 2013) found that HSPG-mediated conformational changes in L1 and L2 allowed the release of the virus from primary binding and allowing secondary receptor engagements in the absence of HS shedding.*
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> *(5) The well-described capsid processing events by KLK8 and furin need to be mechanistically linked to the proposed model. Does inhibition of either of these cleavages prevent engagement with CD151?*
>
> *The authors need to consider an explanation for these discrepancies.*

That PsVs carry HS-cleavage products doesńt imply that HS cleavage is sufficient or required for infection. Therefore, we do not view our data as being in conflict with the priming model. In fact, our observations are compatible with aspects of both the shedding and the priming model.

Yet, we acknowledge that the study would gain importance by directly testing the priming model within our experimental system. As requested by the referee, we will discuss the above papers, and further plan to test KLK8 and furin inhibitors.

> *Other issues:*
>
> *(1) Line 110-111. The statement about PsVs in the ECM being too far away from the cell surface to make physical contact with the cell surface entry receptors is confusing. ECM binding has not been shown to be an obligatory step for in vitro infection.*

Not obligatory, but strongly supportive (Bienkowska-Haba et al., Plos Path., 2018; Surviladze et al., J. Gen. Viro., 2015). As recently published by the Sapp lab (Bienkowska-Haba et al., Plos Path., 2018), Direct binding of HPV16 to primary keratinocytes yields very inefficient infection rates for unknown reasons. Moreover, the paper shows that HaCaT cell ECM binding of PsVs increases the infection of NHEK by 10-fold and of HFK by almost 50-fold.

> *This idea is referred to again on lines 158-159 and 199. The claim (line 158) that PsV does not interact with the cell within an hour needs to be demonstrated experimentally and seems at odds with multiple laboratories' data. PsV has been shown to directly interact with HSPG on the cell surface in addition to the ECM. Why are these PsVs not detected?*

We do not question that in many cellular systems PsVs interact with heparan sulfate proteoglycans (HSPGs) present on the cell surface, or both on the cell surface and the ECM. We stated in the manuscript on line 59 While in cell culture virions bind to HS of the cell surface and the ECM, it has been suggested that in vivo they bind predominantly to HS of the extracellular basement membrane (Day and Schelhaas, 2014; Kines et al., 2009; Schiller et al., 2010).´

Moreover, we ourselves detect these PsVs, for example, in Figure 5A (CytD, 0 min time point), a handful of PsVs localize to the cell body area. However, the large majority overlaps with the strong HS staining at the cell periphery, likely the ECM. An accurate quantification of the fractions of PsVs bound to the ECM/cell body is for the following reasons very difficult. First, the ECM PsVs are very dense and therefore not microscopically resolved into single PsVs, at least not completely (see Figure 1C; the high intensity spots are non-resolved PsVs, please see our discussion on line 148 - 152). For this reason, by just counting spots we strongly underestimate the ECM PsVs versus the cell body PsVs. Second, with the available immunostainings we cannot exactly delineate the ECM from the cell body. In particular, at the cell border region (for example see Figure 4B) we often observe PsV accumulations. Assigning these ćell border region PsVs´entirely to the cell body fraction, a preliminary analysis (correcting for the limitation of non-resolved ECM PsVs) suggests that about a quarter of the PsVs bind to the cell body. On the other hand, assigning them to the ECM, the cell body fraction would be much below 10%. Third, we observe that in regions devoid of ECM and cells PsVs apparently adhere unspecifically to the glass-coverslip. This suggests that some of the cell body PsVs are just unspecific background. Subtraction of a background PsV density from the ECM and cell body PsV density will reduce relatively more the cell body PsVs, and consequently decreases the fraction of cell body PsVs even more.

Moreover, in the course of the project we wondered whether at the basolateral membrane there are not many binding sites anyway. To address this question, in an unpublished experiment, we detached HaCaT cells with trypsin, incubated them with PsVs, and then allowed reattachment to assess the binding in suspension. We detected minimal to no binding, which, however, could also result from apical membrane adherence to the coverslip or trypsin-mediated cleavage of HSPGs. As suggested by the reviewing editor, we agree that repeating this experiment using EDTA for detachment—thus preserving HSPGs—would offer more definitive insight into binding efficiency in the absence of accessibility constraints. In summary, the reason why in our cellular system most PsVs do not bind to the cell surface could be a combination of several factors:

(1) The primary binding partners are more abundant in the ECM and the polarized HaCaT cells secrete more ECM when compared to other cultured cells used to study HPV infection. This promotes ECM binding.

(2) In the polarized HaCaT cells, the apical membrane is largely devoid of syndecan-1, CD151 and Itga6, wherefore PsVs infect the cell via the basolateral membrane. However, the accessibility to the basolateral membrane is restricted, PsVs must diffuse through a narrow

slit between the glass coverslip and the attached cell to reach HS on the cell surface. This limits cell surface binding.

(3) If HaCaT cells secrete large amounts of ECM, the may become depleted from cell surface HS. As outlined above, we will try to find out how many PsVs bind to the basolateral membrane in the absence of restricted accessibility. If it turns out that HaCaT cells have not many binding sites anyway, this would additionally promote binding to the ECM.

The outcome of the above issues, and how we will mention them in the revised version of the manuscript, is open. In any case, we would like to point out that PsVs bound to the cell body do not weaken our main conclusion. Still, we recognize that this point merits attention and plan several modifications of the manuscript. We did already, but now we will mention more explicitly that PsVs have been shown to directly interact with HSPG on the cell surface, in addition to the ECM, but that it also has been shown that the ECM strongly supports infection in NHEK and HFK (Bienkowska-Haba et al., Plos Path., 2018). The following is a draft version of a paragraph we plan to incorporate, explaining the above issue and why we used in our experiments HaCaT cells:

In vitro, PsVs bind to both the cell surface and the ECM, as has been widely documented. In vivo, however, it has been proposed that initial binding occurs predominantly to the basement membrane ECM, rather than directly to the cell surface (Day and Schelhaas, 2014; Kines et al., 2009; Schiller et al., 2010). This distinction reinforces the physiological relevance of ECM-bound particles in the early steps of HPV infection. Support for a functional role of ECM-mediated entry comes from a study showing that PsV binding to ECM derived from HaCaT cells significantly enhances infection of primary keratinocytes (Bienkowska-Haba et al., 2018). For these reasons, we specifically chose polarized HaCaT cells as a model system. These cells secrete abundant ECM from which the cells readily collect bound PsVs. On the other hand, the polarization limits the access of PsVs to basolateral receptors such as CD151 and Itgα6, and also cell body resident Syndecan-1, the most abundant HSPG in keratinocytes (Rapraeger et al., 1986; Hayashi et al., 1987; Kim et al., 1994). Hence, as polarization limits direct cell surface accessibility it biases binding toward the ECM, that in this culture system is abundant. Hence, in the HaCaT cell culture system, like probably in vivo, PsVs cannot circumvent binding to the ECM what they can do in unpolarized cell cultures that may not even secrete significant amounts of ECM. Altogether, this experimental situation closely mimics the in vivo situation where PsVs bind preferentially to the ECM (Day and Schelhaas, 2014; Kines et al., 2009; Schiller et al., 2010).´

We appreciate the reviewer’s input and believe these additions will strengthen the manuscript with regard to the relevance of the used cellular model system.

> *(2) The experiments shown in Figure 5 need to be better controlled. Why is there no HS staining of the cell surface at the early timepoints? This antibody has been shown to recognize N-sulfated glucosamine residues on HS and, therefore, detects HSPG on the ECM and cell surface.*

We have shown all images at the same adjustments of brightness and contrast. As the staining at the periphery is stronger, the impression is given that the cell surface is not stained, although there is some staining. Specific staining is documented in Figure 5D, showing the PCC between PsVs and HS only of the cell body. If there was no HS staining, the PCC would be zero, which is not the case. Yet, it is lower when compared to the PCC measured at the cell border region, with more strongly stained HS.

We will provide images at different contrast and brightness adjustments enabling the reader to see the staining on the cell surface. We will provide also more overview images to illustrate the strong variability of the HS staining between cells.

> *Therefore, the conclusion that this confirms HS coating of PsV during release from the ECM (line 430-431) is unfounded. How do the authors distinguish between "HS-coated virions" and HSPG-associated virions?*

The HS intensity transiently increases on the cell body (Fig. 5D) only after releasing a cohort of PsVs, which can be only explained by PsVs that carry HS from the ECM to the cell body. However, the effect is not significant. Using the antibody 3G10 detecting the HS neoepitope (see the referees' suggestion below) we will reanalyze this point. This should help clarifying the issue.

> *It is difficult to comprehend how the addition of 50 vge/cell of PsV could cause such a global change in HS levels.*

The distribution of bound PsVs largely varies between cells. Some areas are covered with essentially confluent cells, to which hardly any PsVs are bound, because accessing the basolateral membrane of confluent cells is nearly impossible, and PsVs do not bind to the exposed apical membrane. This is different in cultures of unpolarized cells where we expect that PsVs distribute more equally over cells.

This means that in our experiments the vge/cell is not a suitable parameter for relating the magnitude of an effect to a defined number of PsVs. In the ECM, the PsV density is very high, enabling one cell to collect several hundred PsVs, much more than expected from the 50 vge/cell. We will point this out in the revised version.

> *The claim that the HS levels are decreased in the non-cytochalasin-treated cells due to PsV-induced shedding needs to be demonstrated.*

We did not claim that PsVs induce shedding, we rather believe they just take shedded HS with them. Without PsVs, the shedded HS likely remains in the ECM or is washed out very slowly.

> *If HS is actually shed, staining of the cell periphery could increase with the antibody 3G10, which detects the HS neoepitope created following heparinase cleavage.*

As outlined above, we plan to test the suggested antibody 3G10. We also plan to repeat the 0 min time point (with and without PsVs, with and without CytD) to find out whether in the PsV absence the HS intensity (at 0 min) is unchanged between control and CytD.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *Massenberg and colleagues aimed to understand how Human papillomavirus particles that bind to the extracellular matrix (ECM) transfer to the cell body for later uptake, entry, and infection. The binding to ECM is key for getting close to the virus's host cell (basal keratinocytes) after a wounding scenario for later infection in a mouse vaginal challenge model, indicating that this is an important question in the field.*
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> *Strengths:*
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> *The authors take on a conceptually interesting and potentially very important question to understand how initial infection occurs in vivo. The authors confirm previous work that actin-based processes contribute to virus transport to the cell body. The superresolution microscopy methods and data collection are state-of-the art and provide an interesting new way of analysing the interaction with host cell proteins on the cell surface in certain*

> *infection scenarios. The proposed hypothesis is interesting and, if substantiated, could significantly advance the field.*
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> *Weaknesses:*
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> *As a study design, the authors use infection of HaCaT keratinocytes, and follow virus localisation with and without inhibition of actin polymerisation by cytochalasin D (cytoD) to analyse transfer of virions from the ECM to the cell by filopodial structures using important cellular proteins for cell entry as markers.*
>
> *First, the data is mostly descriptive besides the use of cytoD, and does not test the main claim of their model, in which virions that are still bound to heparan sulfate proteoglycans are transferred by binding to tetraspanins along filopodia to the cell body.*

The study identifies a rapid translocation step from the ECM to the cell body. We have no data that demonstrates a physical interaction between PsVs and CD151. In the model figure, we draw CD151 as part of the secondary receptor complex. We are sorry for having raised the impression that PsVs would bind directly to CD151 and will rephrase the respective section.

> *Second, using cytoD is a rather broad treatment that not only affects actin retrograde flow, but also virus endocytosis and further vesicular transport in cells, including exocytosis. Inhibition of myosin II, e.g., by blebbistatin, would have been a better choice as it, for instance, does not interfere with endocytosis of the virus.*

We agree, and plan to test whether blebbistatin is equally efficient in blocking the transfer.

> *Third, the authors aim to study transfer from ECM to the cell body and the effects thereof. However, there are substantial, if not the majority of, viruses that bind to the cell body compared to ECM-bound viruses in close vicinity to the cells.*

We agree that in multiple cell culture systems viruses bind preferentially to the cell directly. But we respectfully disagree with the assertion that the majority of PsVs bind to the cell body of HaCaT keratinocytes. As noted above (e.g., Figure 5A, CytD, 0 min), only a small fraction of PsVs localize to the cell body, whereas the vast majority overlap with intense HS staining at the cell periphery, consistent with ECM association, as the accessibility to the basolateral expressed HSPG is limited (see above). Based on quantitative estimation from multiple images, ECM-bound PsVs largely outnumber cell-bound particles (see above). These features make HaCaT cells a suitable in vitro model for mimicking in vivo conditions, where HPV has been proposed to bind predominantly to the basement membrane ECM rather than the cell surface (Day and Schelhaas, 2014; Kines et al., 2009; Schiller et al., 2010) which also strongly enhances infection of primary keratinocytes in vitro (Bienkowska-Haba et al., 2018).

Thus, we believe our system appropriately models the physiologically relevant scenario of ECM-to-cell transfer, and the observed predominance of ECM binding supports the validity of our experimental focus.

> *This is in part obscured by the small subcellular regions of interest that are imaged by STED microscopy, or by the use of plasma membrane sheets. As a consequence, the obtained data from time point experiments is skewed, and remains for the most part unconvincing due to the fact that the origin of virions in time and space cannot be taken into account. This is particularly important when interpreting association with HS, the tetraspanin CD151, and integral alpha 6, as the low degree of association could originate from cell-bound and ECM-transferred virions alike.*

As stated above, we observe massive binding of PsVs to the ECM, in contrast to very few PsVs that diffuse beneath the basolateral membrane of the polarized HaCaT cells and do bind directly to the cell surface (or maybe they are simply trapped between glass and basolateral membrane). PsVs are not expected to bind to the apical membrane that is depleted from CD151 and Itga6. In other cellular systems, cells may hardly secrete ECM, are not polarized, and do not adhere so tightly to the substrate. In other cultures, where virions can easily circumvent ECM binding, the large majority of PsVs will likely bind directly to the cell surface.

As outlined above, in order to quantify PsVs that can bind without restricted accessibility, we plan to detach HaCaT cells by EDTA from the substrate, incubate them with PsVs, and let them adhere again (please see above).

No matter what is the outcome, the fraction of PsVs that binds directly to the cell surface does not weaken our conclusion that we have identified a very fast and efficient transfer step from the ECM to the cell body.

> *Fourth, the use of fixed images in a time course series also does not allow for understanding the issue of a potential contribution of cell membrane retraction upon cytoD treatment due to destabilisation of cortical actin. Or, of cell spreading upon cytoD washout.*

If blebbistatin works as expected, we can safely conclude that we observe the very same process as described in Scheelhas et al., PLoS Pathogens, 2008, showing that the PsVs migrate by retrograde transport to the cell surface and not that the cell spreads out and by this reaches the PsVs.

> *The microscopic analysis uses an extension of a plasma membrane stain as a marker for ECM-bound virions, which may introduce a bias and skew the analysis.*

Our plasma membrane stain does not stain the ECM. Please see Figure 1. The stain is actually used to distinguish the cell body from the ECM area.

> *Fifth, while the use of randomisation during image analysis is highly recommended to establish significance (flipping), it should be done using only ROIs that have a similar density of objects for which correlations are being established.*

We agree that the way of how randomization is done is very important. Regarding the association of PsVs with CD151 and HS, based on flipped images, we generated a calibration curve used for the correction of random background. For details, please see Supplementary Figures 3 and 5.

> *For instance, if one flips an image with half of the image showing the cell body, and half of the image ECM, it is clear that association with cell membrane structures will only be significant in the original. I am rather convinced that using randomisation only on the plasma membrane ROIs will not establish any clear significance of the correlating signals.*

Figure 5D shows the PCC specifically of the cell body. In flipped images (not shown in the manuscript for clarity, but can be added) we obtain a PCC of around zero. For CytD, the flipped images always have a significantly lower PCC compared to the original images. In the control, the PCC of the flipped images are significantly lower only for the 30 min and 60 min time point. The non-significance of the 0 min and 180 min time point is due to low PCCs also in the original images.

> *Also, there should be a higher n for the measurements.*

One n is the average of 15 cells. We realize that with n = 3 we find significant effects only if the effect is very strong or moderate with very low variance.

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